

Lab 14 – Terraform + Ansible

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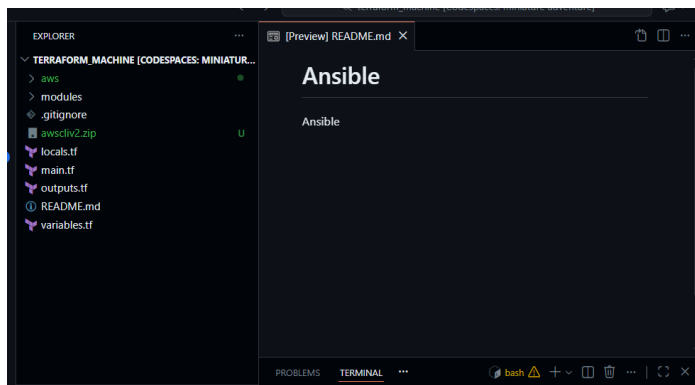
📌 Task 0 – Lab Setup (Codespace & GH CLI)

Tasks

- Open GitHub Codespace
- Verify AWS CLI, Terraform, Ansible
- Verify AWS credentials

Screenshots

- task0_codespace_open.png



- task0_env_check.png

```
@toobashafique065 → /workspaces/terraform_machine (main) $ terraform --version
Terraform v1.14.3
on linux_amd64
@toobashafique065 → /workspaces/terraform_machine (main) $ aws --version
aws-cli/2.32.31 Python/3.13.11 Linux/6.8.0-1030-azure exe/x86_64.ubuntu.24
```

- task0_aws_config.png

```
@toobashafique065 → /workspaces/terraform_machine (main) $ aws sts get-caller-identity
{
  "UserId": "AIDAXZSFF03LFN7RJYPM6",
  "Account": "535942231766",
  "Arn": "arn:aws:iam::535942231766:user/Lab10user"
}
```

📌 Task 1 – Clone Terraform+Ansible Repo & Initial Terraform Apply

Tasks

- Clone Terraform repository
- Generate SSH key pair
- Create terraform.tfvars
- Initialize and apply Terraform (2 instances)
- Verify Terraform outputs

Screenshots

- task1_ssh_keygen_before.png

```
@toobashafique065 [ ] /workspaces/terraform_machine (main) $ ls ~/.ssh
authorized_keys
```

- task1_ssh_keygen.png

```
@toobashafique065 [ ] /workspaces/terraform_machine (main) $ ssh-keygen -t ed25519 -f ~/.ssh/id_ed25519 -N ""
Generating public/private ed25519 key pair.
Your identification has been saved in /home/codespace/.ssh/id_ed25519
Your public key has been saved in /home/codespace/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:LIFd9md/HCH7ivj72+0/miWwrxU6YBG0FBAGxsyleMg codespace@codespaces-1ed512
The key's randomart image is:
+--[ED25519 256]--+
|      +O++.. . . |
|      o.*o.+ . o .|
|      . + + .o.+o. .|
|      E o + .O....|
|      . . S o .oo |
|      . o o +.. |
|      . . B o |
|      . . O.. |
|      oo*++* |
+-----[SHA256]-----+
```

- task1_ssh_keygen_after.png

```
@toobashafique065 [ ] /workspaces/terraform_machine (main) $ ls -la ~/.ssh
total 24
drwxr-xr-x 2 codespace codespace 4096 Jan  8 16:34 .
drwxr-x--- 1 codespace codespace 4096 Jan  8 16:29 ..
-rw-r--r-- 1 codespace codespace 191 Jan  8 16:34 authorized_keys
-rw----- 1 codespace codespace 419 Jan  8 16:34 id_ed25519
-rw-r--r-- 1 codespace codespace 109 Jan  8 16:34 id_ed25519.pub
```

- task1_terraform_tfvars_created.png

```
@toobashafique065 [ ] /workspaces/terraform_machine (main) $ touch terraform.tfvars
@toobashafique065 [ ] /workspaces/terraform_machine (main) $ ls -la terraform.tfvars
-rw-rw-rw- 1 codespace codespace 0 Jan  8 16:40 terraform.tfvars
```

- task1_terraform_tfvars.png

```
vpc_cidr_block = "10.0.0.0/16"
subnet_cidr_block = "10.0.10.0/24"
availability_zone = "me-central-1a"
env_prefix = "dev"
instance_type = "t3.micro"
public_key = "~/.ssh/id_ed25519.pub"
private_key = "~/.ssh/id_ed25519"
```

- task1_terraform_init.png

```
@toobashafique065 [ /workspaces/terraform_machine (main) ] $ terraform init
Initializing the backend...
Initializing modules...
- myapp-subnet in modules/subnet
- myapp-webserver in modules/webserver
Initializing provider plugins...
- Finding latest version of hashicorp/aws...
- Finding latest version of hashicorp/http...
- Installing hashicorp/aws v6.28.0...
- Installed hashicorp/aws v6.28.0 (signed by HashiCorp)
- Installing hashicorp/http v3.5.0...
- Installed hashicorp/http v3.5.0 (signed by HashiCorp)
Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
@toobashafique065 [ /workspaces/terraform_machine (main) ] $
```

- task1_terraform_apply_2_instances.png

```
@toobashafique065 [ /workspaces/terraform_machine (main) ] $ terraform apply -auto-approve
data.http.my_ip: Reading...
data.http.my_ip: Read complete after 0s [id=https://icanhazip.com]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with
the following symbols:
+ create

Terraform will perform the following actions:

# aws_vpc.myapp_vpc will be created
+ resource "aws_vpc" "myapp_vpc" {
+   arn                                = (known after apply)
+   cidr_block                         = "10.0.0.0/16"
+   default_network_acl_id            = (known after apply)
+   default_route_table_id            = (known after apply)
+   default_security_group_id         = (known after apply)
+   dhcp_options_id                   = (known after apply)
+   enable_dns_hostnames               = (known after apply)
+   enable_dns_support                 = true
+   enable_network_address_usage_metrics = (known after apply)
+   id                                 = (known after apply)
+   instance_tenancy                   = "default"
+   ipv6_association_id                = (known after apply)
+   ipv6_cidr_block                    = (known after apply)
+   ipv6_cidr_block_network_border_group = (known after apply)
+   main_route_table_id               = (known after apply)
+   owner_id                           = (known after apply)
+   region                             = "me-central-1"
}
```

- task1_terraform_output_ips.png

```
@toobashafique065 [ /workspaces/terraform_machine (main) ] $ terraform output
webserver_public_ips = [
  "158.252.85.109",
  "3.28.134.247",
]
```

Task 2 – Static Ansible Inventory (2 EC2 Instances)

Tasks

- Install Ansible
- Create static inventory file
- Test Ansible connectivity

Screenshots

- task2_ansible_install.png

```
@toobashafique065 [ ] /workspaces/terraform_machine (main) $ pipx install ansible-core
installed package ansible-core 2.20.1, installed using Python 3.12.1
These apps are now globally available
- ansible
- ansible-config
- ansible-console
- ansible-doc
- ansible-galaxy
- ansible-inventory
- ansible-playbook
- ansible-pull
- ansible-test
- ansible-vault
done! [ ] [ ]
@toobashafique065 [ ] /workspaces/terraform_machine (main) $ ansible --version
ansible [core 2.20.1]
  config file = None
  configured module search path = ['/home/codespace/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/local/py-utils/venvs/ansible-core/lib/python3.12/site-packages/ansible
  ansible collection location = /home/codespace/.ansible/collections:/usr/share/ansible/collections
  executable location = /usr/local/py-utils/bin/ansible
  python version = 3.12.1 (main, Nov 27 2025, 10:47:52) [GCC 13.3.0] (/usr/local/py-utils/venvs/ansible-core/bin/python)
  jinja version = 3.1.6
  pyyaml version = 6.0.3 (with libyaml v0.2.5)
```

- task2_terraform_output_ips.png

```
@toobashafique065 [ ] /workspaces/terraform_machine (main) $ terraform output
webserver_public_ips = [
  "158.252.85.109",
  "3.28.134.247",
]
```

- task2_hosts_created.png

```
@toobashafique065 [ ] /workspaces/terraform_machine (main) $ touch hosts
@toobashafique065 [ ] /workspaces/terraform_machine (main) $ ls -la hosts
-rw-rw-rw- 1 codespace codespace 0 Jan  8 16:50 hosts
```

- task2_hosts_initial.png

```
GNU nano 7.2 hosts
158.252.85.109 ansible_user=ec2-user ansible_ssh_private_key_file=~/.ssh/id_ed25519
3.28.134.247 ansible_user=ec2-user ansible_ssh_private_key_file=~/.ssh/id_ed25519
```

- task2_ansible_ping_initial.png

```
@toobashafique065 [ ] /workspaces/terraform_machine (main) $ ansible all -i hosts -m ping
[ERROR]: Task failed: Failed to connect to the host via ssh: Host key verification failed.
Origin: <adhoc 'ping' task>

{'action': 'ping', 'args': {}, 'timeout': 0, 'async_val': 0, 'poll': 15}
3.28.134.247 | UNREACHABLE! => {
  "changed": false,
  "msg": "Task failed: Failed to connect to the host via ssh: Host key verification failed.",
  "unreachable": true
}
158.252.85.109 | UNREACHABLE! => {
  "changed": false,
  "msg": "Task failed: Failed to connect to the host via ssh: Host key verification failed.",
  "unreachable": true
}
```

- task2_hosts_with_common_args.png

```
158.252.85.109 ansible_user=ec2-user ansible_ssh_private_key_file=~/.ssh/id_ed25519 ansible_ssh_common_args='-o StrictHostKeyChecking=no'
3.28.134.247 ansible_user=ec2-user ansible_ssh_private_key_file=~/.ssh/id_ed25519 ansible_ssh_common_args='-o StrictHostKeyChecking=no'
```

- task2_ansible_ping_success.png

```
@toobashafique065 [ /workspaces/terraform_machine (main) ] $ ansible all -i hosts -m ping
[WARNING]: Host '3.28.134.247' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.20/reference_appendices/interpreter_discovery.html for more information.
3.28.134.247 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.9"
  },
  "changed": false,
  "ping": "pong"
}
[WARNING]: Host '158.252.85.109' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.20/reference_appendices/interpreter_discovery.html for more information.
158.252.85.109 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.9"
  },
  "changed": false,
  "ping": "pong"
}
```

Task 3 – Scale to 3 Instances & Group Inventory

Tasks

- Update Terraform count to 3
- Apply Terraform
- Create grouped Ansible inventory
- Test groups and individual hosts

Screenshots

- task3_main_tf_count_3.png

```
module "myapp-webserver" {
  source = "../modules/webserver"
  env_prefix = var.env_prefix
  instance_type = var.instance_type
  availability_zone = var.availability_zone
  public_key = var.public_key
  my_ip = local.my_ip
  vpc_id = aws_vpc.myapp_vpc.id
  subnet_id = module.myapp-subnet.subnet.id

  # Loop count
  count = 3
  # Use count.index to differentiate instances
  instance_suffix = count.index
}
```

- task3_terraform_apply_3_instances.png

```

@toobashafique065 @ /workspaces/terraform_machine (main) $ terraform apply -auto-approve
data.http.my_ip: Reading...
data.http.my_ip: Read complete after 0s [id=https://icanhazip.com]
module.myapp-webserver[0].aws_key_pair.ssh-key: Refreshing state... [id=dev-serverkey-0]
aws_vpc.myapp_vpc: Refreshing state... [id=vpc-0e89fb9c7b6c7527b]
module.myapp-webserver[1].aws_key_pair.ssh-key: Refreshing state... [id=dev-serverkey-1]
module.myapp-subnet.aws_subnet.myapp_subnet_1: Refreshing state... [id=subnet-07ff526267785d880]
module.myapp-subnet.aws_internet_gateway.myapp_igw: Refreshing state... [id=igw-0232fd16d032b7629]
module.myapp-webserver[0].aws_security_group.web_sg: Refreshing state... [id=sg-00e549d943c197cb4]
module.myapp-webserver[1].aws_security_group.web_sg: Refreshing state... [id=sg-07d68ddc68016f08e]
module.myapp-subnet.aws_default_route_table.main_rt: Refreshing state... [id=rtb-0d413420cf62f26af]
module.myapp-webserver[1].aws_instance.myapp-server: Refreshing state... [id=i-0d40891c2b8f7eb23]
module.myapp-webserver[0].aws_instance.myapp-server: Refreshing state... [id=i-03205fae6da402e84]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with
the following symbols:
+ create

Terraform will perform the following actions:

# module.myapp-webserver[2].aws_instance.myapp-server will be created
+ resource "aws_instance" "myapp-server" {
  + ami                    = "ami-05524d6658fcf35b6"
  + arm                   = (known after apply)
  + associate_public_ip_address = true
  + availability_zone      = "me-central-1a"
  + disable_api_stop       = (known after apply)
  + disable_api_termination = (known after apply)
  + ebs_optimized          = (known after apply)
  + enable_primary_ipv6     = (known after apply)
}

```

- task3_terraform_output_3_ips.png

```

@toobashafique065 @ /workspaces/terraform_machine (main) $ terraform output
webserver_public_ips = [
  "158.252.85.109",
  "3.28.134.247",
  "3.28.184.37",
]

```

- task3_hosts_grouped.png

```

[ec2]
158.252.85.109
3.28.134.247

[ec2:vars]
ansible_user=ec2-user
ansible_ssh_private_key_file=~/.ssh/id_ed25519
ansible_ssh_common_args='-o StrictHostKeyChecking=no'

[droplet]
3.28.184.37

[droplet:vars]
ansible_user=ec2-user
ansible_ssh_private_key_file=~/.ssh/id_ed25519
ansible_ssh_common_args='-o StrictHostKeyChecking=no'

```

- task3_ansible_ec2_ping.png

```

@toobashafique065 @ /workspaces/terraform_machine (main) $ ansible ec2 -i hosts -m ping
[WARNING]: Host '3.28.134.247' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installati
on of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansi
ble-core/2.20/reference_appendices/interpreter_discovery.html for more information.
3.28.134.247 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.9"
  },
  "changed": false,
  "ping": "pong"
}
[WARNING]: Host '158.252.85.109' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installa
tion of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ans
ible-core/2.20/reference_appendices/interpreter_discovery.html for more information.
158.252.85.109 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.9"
  },
  "changed": false,
  "ping": "pong"
}

```

- task3_ansible_single_ip_ping.png

```
@toobashafique065 [ ] /workspaces/terraform_machine (main) $ ansible 158.252.85.109 -i hosts -m ping
[WARNING]: Host '158.252.85.109' is using the discovered Python interpreter at '/usr/bin/python3.9', but future
installation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.20/reference_appendices/interpreter_discovery.html for more information.
158.252.85.109 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.9"
  },
  "changed": false,
  "ping": "pong"
}
```

- task3_ansible_droplet_ping.png

```
@toobashafique065 [ ] /workspaces/terraform_machine (main) $ ansible droplet -i hosts -m ping
[WARNING]: Host '3.28.184.37' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.20/reference_appendices/interpreter_discovery.html for more information.
3.28.184.37 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.9"
  },
  "changed": false,
  "ping": "pong"
}
```

- task3_ansible_all_ping.png

```
@toobashafique065 [ ] /workspaces/terraform_machine (main) $ ansible all -i hosts -m ping
[WARNING]: Host '3.28.184.37' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.20/reference_appendices/interpreter_discovery.html for more information.
3.28.184.37 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.9"
  },
  "changed": false,
  "ping": "pong"
}
[WARNING]: Host '158.252.85.109' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.20/reference_appendices/interpreter_discovery.html for more information.
158.252.85.109 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.9"
  },
  "changed": false,
  "ping": "pong"
}
[WARNING]: Host '3.28.134.247' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.20/reference_appendices/interpreter_discovery.html for more information.
3.28.134.247 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.9"
  },
  "changed": false,
  "ping": "pong"
}
```

📌 Task 4 – Global Ansible Config & First Nginx Playbook

Tasks

- Create global ~/.ansible.cfg
- Remove SSH common args from hosts
- Create nginx playbook
- Deploy nginx on EC2 & droplet

Screenshots

- task4_global_nginx_cfg.png

```
[default]
host_key_checking = False
interpreter_python = /usr/bin/python3
```

- task4_hosts_without_common_args.png

```
[ec2]d 17 lines ]
158.252.85.109
3.28.134.247

[ec2:vars]
ansible_user=ec2-user
ansible_ssh_private_key_file=~/.ssh/id_ed25519

[droplet]
3.28.184.37

[droplet:vars]
ansible_user=ec2-user
ansible_ssh_private_key_file=~/.ssh/id_ed25519
```

- task4_ansible_ping_after_cfg.png

```
@toobashafique065 /workspaces/terraform_machine (main) $ ansible all -i hosts -m ping
[WARNING]: Host '3.28.134.247' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.20/reference_appendices/interpreter_discovery.html for more information.
3.28.134.247 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.9"
  },
  "changed": false,
  "ping": "pong"
}
[WARNING]: Host '158.252.85.109' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.20/reference_appendices/interpreter_discovery.html for more information.
158.252.85.109 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.9"
  },
  "changed": false,
  "ping": "pong"
}
[WARNING]: Host '3.28.184.37' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.20/reference_appendices/interpreter_discovery.html for more information.
3.28.184.37 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.9"
  },
  "changed": false,
  "ping": "pong"
}
```

- task4_my_playbook_created.png

```
@toobashafique065 /workspaces/terraform_machine (main) $ touch my-playbook.yaml
@toobashafique065 /workspaces/terraform_machine (main) $ ls -la my-playbook.yaml
-rw-rw-rw- 1 codespace codespace 0 Jan  9 18:16 my-playbook.yaml
```

- task4_my_playbook_ec2.png


```
GNU nano 7.2
---
- name: Configure nginx web server
  hosts: ec2
  become: true
  tasks:
    - name: install nginx and update cache
      yum:
        name: nginx
        state: present
        update_cache: yes

    - name: start nginx server
      service:
        name: nginx
        state: started
```

- task4_nginx_play_ec2.png

```
@toobashafique065 [ /workspaces/terraform_machine (main) ] $ ansible-playbook -i hosts my-playbook.yaml

PLAY [Configure nginx web server] *****

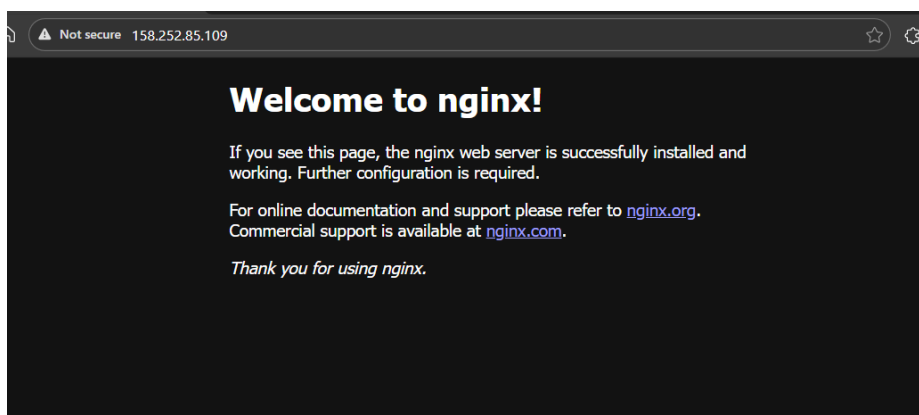
TASK [Gathering Facts] *****
[WARNING]: Host '3.28.134.247' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installati
on of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansi
ble-core/2.20/reference_appendices/interpreter_discovery.html for more information.
ok: [3.28.134.247]
[WARNING]: Host '158.252.85.109' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installa
tion of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/an
sible-core/2.20/reference_appendices/interpreter_discovery.html for more information.
ok: [158.252.85.109]

TASK [install nginx and update cache] *****
changed: [3.28.134.247]
changed: [158.252.85.109]

TASK [start nginx server] *****
changed: [3.28.134.247]
changed: [158.252.85.109]

PLAY RECAP *****
158.252.85.109      : ok=3    changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
3.28.134.247       : ok=3    changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

- task4_nginx_browser_ec2.png



- task4_my_playbook_droplet.png

```

- name: Configure nginx web server
  hosts: droplet
  become: true
  tasks:
    - name: install nginx and update cache
      yum:
        name: nginx
        state: present
        update_cache: yes
    - name: start nginx server
      service:
        name: nginx
        state: started

```

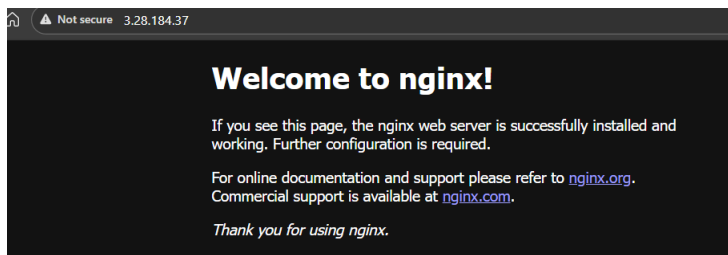
- task4_nginx_play_droplet.png

```

@toobashafique065 [ /workspaces/terraform_machine (main) ] $ ansible-playbook -i hosts my-playbook.yaml
PLAY [Configure nginx web server] *****
TASK [Gathering Facts] *****
[WARNING]: Host '3.28.184.37' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installatio
n of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansib
le-core/2.20/reference_appendices/interpreter_discovery.html for more information.
ok: [3.28.184.37]
TASK [install nginx and update cache] *****
changed: [3.28.184.37]
TASK [start nginx server] *****
changed: [3.28.184.37]
PLAY RECAP *****
3.28.184.37 : ok=3 changed=2 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0

```

- task4_nginx_browser_droplet.png



📌 Task 5 – Single Nginx Target & HTTPS Prerequisites

Tasks

- Create project-level ansible.cfg
- Reduce EC2 count to 1
- Configure nginx + openssl

Screenshots

- task5_project_nginx_cfg_created.png

```

@toobashafique065 [ /workspaces/terraform_machine (main) ] $ touch ansible.cfg
@toobashafique065 [ /workspaces/terraform_machine (main) ] $ ls -la ansible.cfg
-rw-rw-rw- 1 codespace codespace 0 Jan  9 18:26 ansible.cfg

```

- task5_project_nginx_cfg.png

```
GNU nano 7.2
[defaults]
host_key_checking=False
interpreter_python = /usr/bin/python3
```

- task5_main_tf_count_1.png

```
module "myapp-webserver" {
  source = "./modules/webserver"
  env_prefix = var.env_prefix
  instance_type = var.instance_type
  availability_zone = var.availability_zone
  public_key = var.public_key
  my_ip = local.my_ip
  vpc_id = aws_vpc.myapp_vpc.id
  subnet_id = module.myapp-subnet.subnet.id

  # Loop count
  count = 1
}
```

- task5_terraform_apply_one_instance.png

```
@toobashafique065 @ /workspaces/terraform_machine (main) $ terraform apply -auto-approve
module.myapp-webserver[2].aws_key_pair.ssh-key: Refreshing state... [id=dev-serverkey-2]
module.myapp-webserver[0].aws_key_pair.ssh-key: Refreshing state... [id=dev-serverkey-0]
aws_vpc.myapp_vpc: Refreshing state... [id=vpc-0e89fb9c7b6c7527b]
module.myapp-webserver[1].aws_key_pair.ssh-key: Refreshing state... [id=dev-serverkey-1]
module.myapp-subnet.aws_subnet.myapp_subnet_1: Refreshing state... [id=subnet-07ff526267785d880]
module.myapp-webserver[0].aws_security_group.web_sg: Refreshing state... [id=sg-00e549d943c197cb4]
module.myapp-webserver[1].aws_security_group.web_sg: Refreshing state... [id=sg-07d68ddc68016f08e]
module.myapp-webserver[2].aws_security_group.web_sg: Refreshing state... [id=sg-0d09aafe9931dd818]
module.myapp-subnet.aws_internet_gateway.myapp_igw: Refreshing state... [id=igw-0232fd16d032b7629]
module.myapp-webserver[2].aws_instance.myapp-server: Refreshing state... [id=i-0516cee1c0cb17420]
module.myapp-webserver[1].aws_instance.myapp-server: Refreshing state... [id=i-0d40891c2b8f7eb23]
module.myapp-webserver[0].aws_instance.myapp-server: Refreshing state... [id=i-03205fae6da402e84]
module.myapp-subnet.aws_default_route_table.main_rt: Refreshing state... [id=rtb-0d413420cf62f26af]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated
following symbols:
- destroy

Terraform will perform the following actions:

# module.myapp-webserver[1].aws_instance.myapp-server will be destroyed
# (because module.myapp-webserver[1] is not in configuration)
- resource "aws_instance" "myapp-server" {
  - ami                                = "ami-05524d6658fcf35b6" -> null
  - arn                                = "arn:aws:ec2:me-central-1:535942231766:instance/i-0d40891c2b8f7eb23" -> null
  - associate_public_ip_address       = true -> null
  - availability_zone                  = "me-central-1a" -> null
  - disable_api_stop                   = false -> null
  - disable_api_termination            = false -> null
  - ebs_optimized                      = false -> null
```

- task5_terraform_output_single_ip.png

```
@toobashafique065 @ /workspaces/terraform_machine (main) $ terraform output
webserver_public_ips = [
  "158.252.85.109",
]
```

- task5_hosts_nginx_group.png

```
GNU nano 7.2
[nginx]
158.252.85.109
[nginx:vars]
ansible_ssh_private_key_file=~/.ssh/id_ed25519
ansible_user=ec2-user
```

- task5_my_playbook_nginx_group.png

```
---
- name: Configure nginx web server
  hosts: nginx
  become: true
  tasks:
    - name: install nginx and update cache
      yum:
        name: nginx
        state: present
        update_cache: yes

    - name: install openssl
      yum:
        name: openssl
        state: present

    - name: start nginx server
      service:
        name: nginx
        state: started
        enabled: true
```

- task5_nginx_play_nginx_group.png

```
@toobashafique065 @ /workspaces/terraform_machine (main) $ ansible-playbook -i hosts my-playbook.yaml

PLAY [Configure nginx web server] *****

TASK [Gathering Facts] *****
ok: [158.252.85.109]

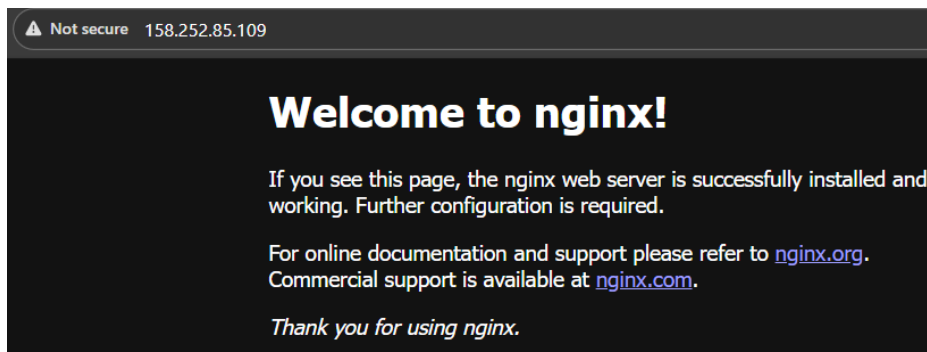
TASK [install nginx and update cache] *****
ok: [158.252.85.109]

TASK [install openssl] *****
ok: [158.252.85.109]

TASK [start nginx server] *****
changed: [158.252.85.109]

PLAY RECAP *****
158.252.85.109 : ok=4  changed=1  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0
```

- task5_nginx_browser_single.png



Task 6 – Ansible Managed SSL Certificates

Tasks

- Generate IMDSv2 token
- Create self-signed SSL certificates
- Verify SSL files on EC2

Screenshots

- task6_my_playbook_ssl_section.png

```
---
- name: Configure nginx web server
  hosts: nginx
  become: true
  tasks:
    - name: install nginx and update cache
      yum:
        name: nginx
        state: present
        update_cache: yes

    - name: install openssl
      yum:
        name: openssl
        state: present

    - name: start nginx server
      service:
        name: nginx
        state: started
        enabled: true

- name: Configure SSL certificates
  hosts: nginx
  become: true
  tasks:
    - name: Create SSL private directory
      file:
        path: /etc/ssl/private
        state: directory
        mode: '0700'
```

- task6_ansible_play_ssl.png

```

root@toobashafique065: /workspaces/terraform_machine (main) $ ansible-playbook -i hosts my-playbook.yaml
[WARNING]: Ansible is being run in a world writable directory (/workspaces/terraform_machine), ignoring it as an ansible
.cfg source. For more information see https://docs.ansible.com/ansible/devel/reference_appendices/config.html#cfg-in-wor
ld-writable-dir

PLAY [Configure nginx web server] *****

TASK [Gathering Facts] *****
[WARNING]: Host '158.252.85.109' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installa
tion of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/an
sible-core/2.20/reference_appendices/interpreter_discovery.html for more information.
ok: [158.252.85.109]

TASK [install nginx and update cache] *****
ok: [158.252.85.109]

TASK [install openssl] *****
ok: [158.252.85.109]

TASK [start nginx server] *****
ok: [158.252.85.109]

PLAY [Configure SSL certificates] *****

TASK [Gathering Facts] *****
ok: [158.252.85.109]

```

- task6_ssl_cert_file.png 158.252.85.109

```

#
~\    #####          Amazon Linux 2023
nnn \   ##### \
      \### |
      \#/     https://aws.amazon.com/linux/amazon-linux-2023
      V~'-'>
      .
      /
      /m/'
Last login: Fri Jan 9 19:16:56 2026 from 4.240.18.226
[ec2-user@ip-10-0-10-74 ~]$ sudo cat /etc/ssl/certs/selfsigned.crt
-----BEGIN CERTIFICATE-----
MIIDQTCCAimgAwIBAgIUf7C1kQDYrSyFckDPmma8PGN50kcwDQYJKoZIhvcNAQEL
BQAQGTEXEMBUGA1UEAAgUMTU4LjI1Mi44NS4xMDkwHhcNMjYwMTA5MTkxNjU2WWhcN
MjcwMTA5MTkxNjU2WjAZMRcwFQYDVQQDDA4xNTguMjUyLjg1LjEwOTCCASIwDQYJ
KoZIhvcNAQEBAQADggEPADCCAQCgGEBAlEfrcrh90mc8II1duRRbJ0pZz2DpkFKuK
Py6JmyQDRRTfcwonW1Nqk5E3d8PCc9uEfztTN3DuCpSrexlbWe02Ay1fhtUuc0Xxf
OJCy+mw47+a4UoylygImun4LFv+IZNuQgiQ17U13ITrfxiOFYwwlWnwQd26NSB0S
YLereb0Wz9KZa7GbmmFfkHG/o36eLc6RgVJ5rv3sMppzWmeQDBztK+AeU5NrZyois
o2tTGziBhzKB9t1VTYAmmWQZ8ylS5VST+RK7MMxpWzj607inAbun1nv+CjRi/eOCp
mINQxllegykUnvJtPHM+1gQ+p/uXvjFVxQFabQOdYa8hsPuyiXuzH4cCAwEAa0B
gdB+MB0GA1UdDgQWBTT64nl3eNFNxUnnUAkJ7PxKmiiTNNzAFBgNVHSMEGDAGwBT6
4nl3eNFNxUnnUAkJ7PxKmiiTNNzAPBgNVHREEDAGHwSe/FVtMAkGA1UdEwQCAAaw
CWYVR0PBAAQDAgWMBMGAGA1UdJQQMMAoGCCsGAQUFBwMBMAoGCCsGAQUFBzABDAQE
A4IBAQA4ogdZvkrFyhKZUAsp1eh2MMWIwqQJ37zKBPgqqEhhUW4Y7SEiuAkuKaE7n
mkNkYmwrVbIzbhgcsizI4fewS23CnzbIZuVkcj4Fc4VqbQ1kyXwS/L0/nXT0ShLyN
EzLIzUKAI2H07ONGKpwIseBkpVo9KZlBTWBpn95dqmpcpYa8lhMZOs96018LEQW
ZZYXwoPEVvhVyp1xbPnbGervVefN7idvhiWkhzu+eLTcVN5MVyThs0Kxu85EDfwDx
lHcw6DzF3QelIKHBVpwK34LvahJhS/H5IuMukcmfbHiTANLPp9DrQ97wYYAlQqn
FWzkL1lrLhkyJ0636YmIEVPm8zo6
-----END CERTIFICATE-----

```

- task6_ssl_key_file.png

```
[ec2-user@ip-10-0-10-74 ~]$ sudo cat /etc/ssl/private/selfsigned.key
-----BEGIN PRIVATE KEY-----
MIIEvgIBADANBgkqhkiG9w0BAQEFAASCBKgwggSkAgEAAoIBAQCCHn63IfdJnPCNX
bKUYdydKwc9g6ZBZLiJ8uiZskAUbX3MKJ1tTapORN3fDwnPbhH87Uzdw7gqUq3sW
1ntNgMpX4bVLnNMV39CQsvpluO/muFKMtcoCJrp+Cxb/iGTbkIikNe1NdyE638Yj
hWM3Vp1qg9ujUgdEmCk3m9Fs/SmWuxm5jBZBxv6N+ni30kYFSea797DKac1pnk
Awc7Svgh10Ta2cqIuaNrUxmYgYcygfbDVU2A3lkGfMi0uVUK/kSuzDMacGY+t04p
wG7p9Z7/go0Yv3jggZiDajS2XoMq1J7ybtXzPtYEPqf711YxVcUBWm0DnWgVibKV
Mo17sx+HAgMBAAECggEAP21HZcB+icoydASYxQjkl3hG1zaIvhULaYTPRrNIsn20
WgEA0Ew6bWFT2/G1HfQbUO+dsm16AP0Udh8eU6Rt9x0PGSVgB3A7AgzMybpbhCaW
Yjf1FYBqHditjhiFn6s3ThrwsxGKNe6+w4jwQJ00cglrYbMpDn6z0uMq7mAmFz76
PJ+ehuDgdWefuYZmpffCsnczESGAn7b6YLnzFuwzhhnCsbPr9Pa295ZDV7c4mAV
vZVxN20VUCz4Iet8QgwXcqzpkU/Dp+kbZbbGgkByV75wUaNL1trssdC/EiU1UrOH
v+zQW9uVa+VGh/sYeRJDsa0qgtf2xbsrFdoMXIFboQKBGQC8YT+fba7Y0ks32TKt
g87zI0waj2xqDRqpjQW4a1WoDHeKP12kkX6tuoC76mmJ0A3VHNmBt9o8UKwUbza0
+BxUTNT8wSA7wAB4wuYx1yU08rQTM0g/32SefS80y/ARd0jyioXE3knaCucMgBsJ
Q9R000wn6gYbWke3Q0mPxPEv3wKBGQC4ToXA9osLuK8S2gI4Keek7v1w0T7xCvvV
aD9SMD6TTszoKv0+fUZEG9mgb2sqXzTw6fq1D3zya+iYehSxcXhXzrupfadJ8tn
7WjMd1wHTR0CJELIXeujDCz09C+SgEa0hJJ/vzsJx3rX4A+Jh72Y6zXaKWTMve96
wG51UNn1WQKBGQCki5mES81HTOEw5k3y98gj7ngCFzd0pywb5jeR8AeA42rjUQ6T
ShdJ7reV7NrE6vdcF3gR66Lv10hAceS0BWeuQhtyU+7gSmoU126aT/Qf8U1ZDz+
tcXw/kdJmXx+W6BXYIXdUcy8wUAKUGMZThFrYH094bgC4rLFDZEI3WJwKBGQCK
kud4F5aocnyXpikN8WmOdMcRsd3PpXod9AcQI/b/ASOjgNAP7o8HwB8gy1lG15X
wOSmVEXSk5W3hxq167XuBdwGcRxCPxhtlJAWIuvAxZcI9DCOLCeumMob9HcMRIxI
PZ9Uk3iV0rm/NAUA18fTAQ51cHgMhMZZWfPnE2oxSQKBGAsQuPV0x0BhpfR01nb
0gAIQI8dvFyAbrnBwe6ak0In/XER1Wc700wV2u0BsR0zSh/wM8Pq8u35Znb0J1Rm
Fb0u0DCSm0CcCS7on0mB/uht9dNpBzYDpvHXv1pnNiz1f91YcQRURax2N0mIstZL
3hUhBP6g37Ym/eq7eu27jUe4
-----END PRIVATE KEY-----
[ec2-user@ip-10-0-10-74 ~]$
```

Task 7 – PHP Frontend Deployment

Tasks

- Create PHP metadata page
- Create Nginx template
- Deploy PHP + HTTPS website

Screenshots

- task7_files_templates_created.png

```
@toobashafique065 [ /workspaces/terraform_machine (main) ] $ cd /workspaces/terraform_machine
@toobashafique065 [ /workspaces/terraform_machine (main) ] $ mkdir -p files templates
@toobashafique065 [ /workspaces/terraform_machine (main) ] $ touch files/index.php
@toobashafique065 [ /workspaces/terraform_machine (main) ] $ touch templates/nginx.conf.j2
@toobashafique065 [ /workspaces/terraform_machine (main) ] $ ls -R
.:
README.md      awscli.v2.zip  locals.tf      my-playbook.yaml  templates      terraform.tfvars
ansible.cfg    aws            hosts          my-playbook.yaml.save  terraform.tfstate  variables.tf
aws            awscli        modules        outputs.tf          terraform.tfstate.backup  {}

./aws:
README.md  THIRD_PARTY_LICENSES  dist  install

./aws/dist:
_awscli.abi3.so      _ruamel_yaml.cpython-313-x86_64-linux-gnu.so  base_library.zip  libffi.so.6  libtinfo.so.5
aws                 awscli                                           dateutil         liblzma.so.5  libuuid.so.1
aws_completer       docutils                                         libpython3.13.so.1.0  libz.so.1
awscli              lib-dynload                                     libreadline.so.6     prompt_toolkit-3.0.51.dist-info
                    libbz2.so.1                                     libsqlite3.so.0       wheel-0.45.1.dist-info

./aws/dist/awscli:
botocore  customizations  data  examples  topics

./aws/dist/awscli/botocore:
cacert.pem  data
```

- task7_index_php_content.png

```

GNU nano 7.2
<?php
// Get hostname
$hostname = gethostname();

// Deployment date
$deployed_date = date("Y-m-d H:i:s");

// Metadata base URL
$metadata_base = "http://169.254.169.254/latest/";

// Function to get IMDSv2 token
function getImdsV2Token() {
    $ch = curl_init("http://169.254.169.254/latest/api/token");
    curl_setopt_array($ch, [
        CURLOPT_RETURNTRANSFER => true,
        CURLOPT_CUSTOMREQUEST => "PUT",
        CURLOPT_HTTPHEADER => [
            "X-aws-ec2-metadata-token-ttl-seconds: 21600"
        ],
        CURLOPT_TIMEOUT => 2
    ]);

    $token = curl_exec($ch);
    curl_close($ch);

    return $token ?: null;
}

// Function to fetch metadata using token
function getMetadata($path, $token) {
    $url = "http://169.254.169.254/latest/meta-data/" . $path;

    $ch = curl_init($url);
    curl_setopt_array($ch, [
        CURLOPT_RETURNTRANSFER => true,
        CURLOPT_HTTPHEADER => [
            "X-aws-ec2-metadata-token: $token"
        ],
        CURLOPT_TIMEOUT => 2
    ]);
}

```

- task7_nginx_conf_template.png

```

user nginx;
worker_processes auto;
error_log /var/log/nginx/error.log notice;
pid /run/nginx.pid;

events {
    worker_connections 1024;
}

http {
    log_format main '$remote_addr - $remote_user [$time_local] "$request" '
        '$status $body_bytes_sent "$http_referer" '
        '"$http_user_agent" "$http_x_forwarded_for"';

    access_log /var/log/nginx/access.log main;

    sendfile on;
    tcp_nopush on;
    keepalive_timeout 65;
    types_hash_max_size 4096;

    include /etc/nginx/mime.types;
    default_type application/octet-stream;

    upstream backend_servers {
        server 158.252.85.109:80;
    }

    server {
        listen 443 ssl;
        server_name {{ server_public_ip }};
        ssl_certificate /etc/ssl/certs/selfsigned.crt;
        ssl_certificate_key /etc/ssl/private/selfsigned.key;

        location / {
            root /usr/share/nginx/html;
            index index.php index.html index.htm;

            location ~ \.php$ {
                include fastcgi_params;

```

- task7_my_playbook_web_deploy.png


```

- name: Deploy Nginx website and configuration files
  hosts: nginx
  become: true
  vars:
    server_public_ip: "{{ ansible_host }}"
  tasks:
    - name: install php-fpm and php-curl
      yum:
        name:
          - php-fpm
          - php-curl
        state: present

    - name: Copy website files
      copy:
        src: files/index.php
        dest: /usr/share/nginx/html/index.php
        owner: nginx
        group: nginx
        mode: '0644'

    - name: Copy nginx.conf template
      template:
        src: templates/nginx.conf.j2
        dest: /etc/nginx/nginx.conf
        owner: root
        group: root
        mode: '0644'

    - name: Restart nginx
      service:
        name: nginx
        state: restarted

    - name: Start and enable php-fpm

```

- task7_ansible_play_web_deploy.png

```

@toobashafique065 @ /workspaces/terraform_machine (main) $ ansible-playbook -i hosts my-playbook.yaml
[WARNING]: Ansible is being run in a world writable directory (/workspaces/terraform_machine), ignoring it as an ansible
.cfg source. For more information see https://docs.ansible.com/ansible/devel/reference_appendices/config.html#cfg-in-wor
ld-writable-dir

PLAY [Configure nginx web server] *****

TASK [Gathering Facts] *****
[WARNING]: Host '158.252.85.109' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installa
tion of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/an
sible-core/2.20/reference_appendices/interpreter_discovery.html for more information.
ok: [158.252.85.109]

TASK [install nginx and update cache] *****
ok: [158.252.85.109]

TASK [install openssl] *****
ok: [158.252.85.109]

TASK [start nginx server] *****
ok: [158.252.85.109]

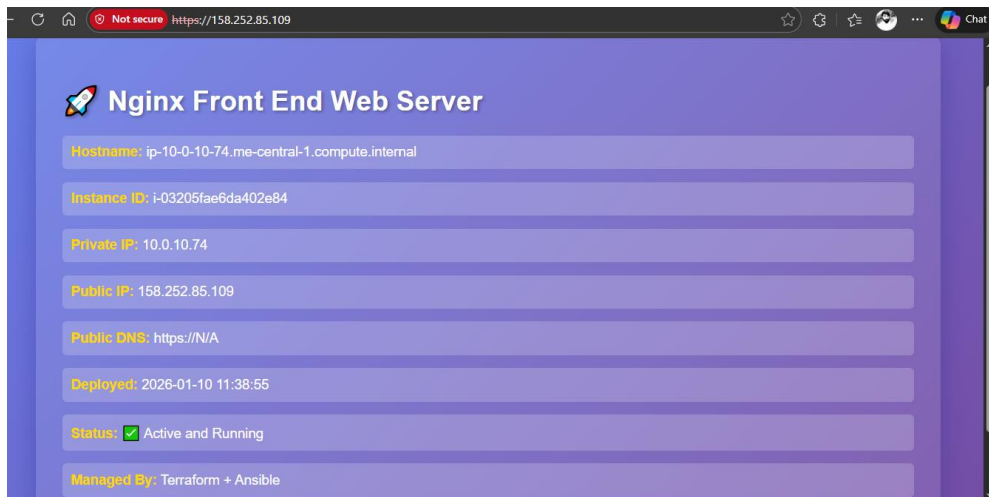
PLAY [Configure SSL certificates] *****

TASK [Gathering Facts] *****
ok: [158.252.85.109]

TASK [Create SSL private directory] *****
ok: [158.252.85.109]

```

- task7_php_https_browser.png



Task 8 – Docker & Docker Compose via Ansible

Tasks

- Destroy old infra
- Create Docker EC2 instance
- Install Docker & Docker Compose

Screenshots

- task8_terraform_destroy_old.png

```
strobach@strobach: ~$ terraform destroy -auto-approve
module.myapp-webserver[0].aws_key_pair.ssh-key: Refreshing state... [id=dev-serverkey-0]
aws_vpc.myapp_vpc: Refreshing state... [id=vpc-0e89fb9c7b6c7527b]
module.myapp-subnet.aws_internet_gateway.myapp_igw: Refreshing state... [id=igw-0232fd16d032b7629]
module.myapp-subnet.aws_subnet.myapp_subnet_1: Refreshing state... [id=subnet-07ff526267785d880]
module.myapp-webserver[0].aws_security_group.web_sg: Refreshing state... [id=sg-00e549d943c197cb4]
module.myapp-subnet.aws_default_route_table.main_rt: Refreshing state... [id=rtb-0d413420cf62f26af]
module.myapp-webserver[0].aws_instance.myapp-server: Refreshing state... [id=i-03205fae6da402e84]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  - destroy

Terraform will perform the following actions:

# aws_vpc.myapp_vpc will be destroyed
- resource "aws_vpc" "myapp_vpc" {
  - arn                                = "arn:aws:ec2:me-central-1:535942231766:vpc/vpc-0e89fb9c7b6c7527b" -> null
  - assign_generated_ipv6_cidr_block   = false -> null
  - cidr_block                         = "10.0.0.0/16" -> null
  - default_network_acl_id            = "acl-0cdccb62e169e82ec" -> null
  - default_route_table_id            = "rtb-0d413420cf62f26af" -> null
  - default_security_group_id         = "sg-06469fcae02526253" -> null
  - dhcp_options_id                   = "dopt-02c0a1e8f3dd284d6" -> null
  - enable_dns_hostnames               = false -> null
  - enable_dns_support                 = true -> null
  - enable_network_address_usage_metrics = false -> null
  - id                                 = "vpc-0e89fb9c7b6c7527b" -> null
  - instance_tenancy                  = "default" -> null
  - ipv6_netmask_length                = 0 -> null
  - main_route_table_id                = "rtb-0d413420cf62f26af" -> null
  - owner_id                           = "535942231766" -> null
  - region                             = "me-central-1" -> null
  - tags                               = {
    -   "Name" = "dev-vpc"
  } -> null
} -> null
tags all
```

- task8_terraform_apply_docker_instance.png

```

@toobashafique065 @ /workspaces/terraform_machine (main) $ terraform apply -auto-approve

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the symbol
+ create

Terraform will perform the following actions:

# aws_vpc.myapp_vpc will be created
+ resource "aws_vpc" "myapp_vpc" {
  + arn                               = (known after apply)
  + cidr_block                        = "10.0.0.0/16"
  + default_network_acl_id           = (known after apply)
  + default_route_table_id           = (known after apply)
  + default_security_group_id        = (known after apply)
  + dhcp_options_id                  = (known after apply)
  + enable_dns_hostnames              = (known after apply)
  + enable_dns_support                = true
  + enable_network_address_usage_metrics = (known after apply)
  + id                               = (known after apply)
  + instance_tenancy                  = "default"
  + ipv6_association_id               = (known after apply)
  + ipv6_cidr_block                   = (known after apply)
  + ipv6_cidr_block_network_border_group = (known after apply)
  + main_route_table_id               = (known after apply)
  + owner_id                          = (known after apply)
  + region                            = "me-central-1"
  + tags                              = {
    + "Name" = "dev-vpc"
  }
  + tags_all                          = {
    + "Name" = "dev-vpc"
  }
}

```

- task8_terraform_output_new_ip.png

```

Outputs:

webserver_public_ips = [
  "158.252.78.82",
]

```

- task8_hosts_docker_servers.png

```

GNU nano 7.2
[docker_servers]
158.252.78.82

[docker_servers:vars]
ansible_ssh_private_key_file=~/.ssh/id_ed25519
ansible_user=ec2-user

```

- task8_my_playbook_docker.png

```

GNU nano 7.2 my-playbook.yml
- name: Configure Docker
  hosts: all
  become: true
  tasks:
    - name: install docker and update cache
      yum:
        name: docker
        state: present
        update_cache: yes

- name: Install Docker Compose
  hosts: all
  become: true
  gather_facts: true
  tasks:
    - name: create docker cli-plugins directory
      file:
        path: /usr/local/lib/docker/cli-plugins
        state: directory
        mode: '0755'

    - name: install docker-compose
      get_url:
        url: https://github.com/docker/compose/releases/latest/download/docker-compose-linux-{{ lookup('pipe', 'uname -m') }}
        dest: /usr/local/lib/docker/cli-plugins/docker-compose
        mode: +x

    - name: View architecture of the system
      debug:
        msg: "System architecture of {{ inventory_hostname }} is {{ ansible_facts['architecture'] }}"

    - name: Alternate method to view architecture of the system
      debug:
        msg: "System architecture of {{ inventory_hostname }} is {{ lookup('pipe', 'uname -m') }}"

    - name: restart docker service
      service:
        name: docker
        state: restarted

```

- task8_ansible_play_docker.png

- task8_docker_ps_remote.png

Task 9 – Gitea Docker Stack & Security Group

- Add user to docker group
- Deploy Gitea + Postgres
- Open port 3000
- Access Gitea UI

- task9_my_playbook_add_user_to_docker.png

```

- name: Adding user to docker group
  hosts: all
  become: true
  vars_files:
    - project-vars.yaml
  tasks:
    - name: add user to docker group
      user:
        name: "{{ normal_user }}"
        groups: docker
        append: yes

    - name: reconnect to apply group changes
      meta: reset_connection

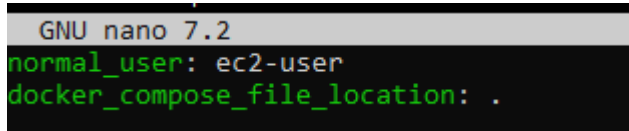
    - name: verify docker access
      command: docker ps
      register: docker_ps
      changed_when: false

    - name: display docker ps output
      debug:
        var: docker_ps.stdout

    - name: fail if docker is not accessible
      fail:
        msg: "Docker is not accessible on this host"
        when: docker_ps.rc != 0

```

- task9_project_vars.png



```

GNU nano 7.2
normal_user: ec2-user
docker_compose_file_location: .

```

- task9_my_playbook_deploy_containers.png

```

- name: Deploy Docker Containers
  hosts: all
  become: true
  user: "{{ normal_user }}"
  vars_files:
    - project-vars.yaml
  tasks:
    - name: check if docker-compose file exists
      stat:
        path: /home/{{ normal_user }}/compose.yaml
      register: compose_file

    - name: copy docker-compose file
      copy:
        src: "{{ docker_compose_file_location }}/compose.yaml"
        dest: /home/{{ normal_user }}/compose.yaml
        mode: '0644'
      when: not compose_file.stat.exists

    - name: deploy containers using docker-compose
      command: docker compose up -d
      args:
        chdir: /home/{{ normal_user }}
      register: compose_result
      changed_when: "'Creating' in compose_result.stdout or 'Recreating' in compose_result.stdout"

```

- task9_compose_yaml.png

```

GNU nano 7.2
services:
  gitea:
    image: gitea/gitea:latest
    container_name: gitea
    environment:
      - DB_TYPE=postgres
      - DB_HOST=db:5432
      - DB_NAME=gitea
      - DB_USER=gitea
      - DB_PASSWD=gitea
    restart: always
    volumes:
      - gitea:/data
    ports:
      - 3000:3000
    networks:
      - webnet

  db:
    image: postgres:alpine
    container_name: gitea_db
    environment:
      - POSTGRES_USER=gitea
      - POSTGRES_PASSWORD=gitea
      - POSTGRES_DB=gitea
    restart: always
    volumes:
      - gitea_postgres:/var/lib/postgresql/data
    expose:
      - 5432
    networks:
      - webnet

volumes:
  gitea_postgres:
    name: gitea_postgres
  gitea:
    name: gitea

networks:

```

- task9_ansible_play_gitea.png

```

root@terraform-machine: /workspaces/terraform_machine (main) $ ansible-playbook -i hosts my-playbook.yaml
[WARNING]: Ansible is being run in a world writable directory (/workspaces/terraform_machine), ignoring it as an ansible.cfg source. For more informat
cs.ansible.com/ansible/develop/reference_appendices/config.html#cfg-in-world-writable-dir

PLAY [Configure Docker] *****
TASK [Gathering Facts] *****
[WARNING]: Host 158.252.78.82 is using the discovered Python interpreter at /usr/bin/python3.9, but future installation of another Python interp
different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.20/reference_appendices/interpreter_discovery.html for more inform
ok: [158.252.78.82]

TASK [Install docker and update cache] *****
ok: [158.252.78.82]

PLAY [Install Docker Compose] *****
TASK [Gathering Facts] *****
ok: [158.252.78.82]

TASK [create docker cli-plugins directory] *****
ok: [158.252.78.82]

TASK [Install docker-compose] *****
ok: [158.252.78.82]

TASK [View architecture of the system] *****
ok: [158.252.78.82] => {
  "msg": "System architecture of 158.252.78.82 is x86_64"
}

TASK [Alternate method to view architecture of the system] *****
ok: [158.252.78.82] => {
  "msg": "System architecture of 158.252.78.82 is x86_64"
}

```

- task9_sg_ingress_3000.png

```

}
ingress {
  from_port    = 3000
  to_port      = 3000
  protocol     = "tcp"
  cidr_blocks  = ["0.0.0.0/0"]
}

```

- task9_terraform_apply_sg_3000.png

```

@toobashafique065 [ ] /workspaces/terraform_machine (main) $ terraform apply -auto-approve
module.myapp-webserver[0].aws_key_pair.ssh-key: Refreshing state... [id=dev-serverkey-0]
aws_vpc.myapp_vpc: Refreshing state... [id=vpc-0f313f58fd0996cd0]
module.myapp-subnet.aws_internet_gateway.myapp_igw: Refreshing state... [id=igw-0a86095ed97fd33d4]
module.myapp-subnet.aws_subnet.myapp_subnet_1: Refreshing state... [id=subnet-0a454ec6f8e2a1ce5]
module.myapp-webserver[0].aws_security_group.web_sg: Refreshing state... [id=sg-0bf285c4870ec7dff]
module.myapp-subnet.aws_default_route_table.main_rt: Refreshing state... [id=rtb-0d1f69a2073f9fd17]
module.myapp-webserver[0].aws_instance.myapp-server: Refreshing state... [id=i-01d7b80f722d4f566]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
~ update in-place

Terraform will perform the following actions:

# module.myapp-webserver[0].aws_security_group.web_sg will be updated in-place
~ resource "aws_security_group" "web_sg" {
  id          = "sg-0bf285c4870ec7dff"
  ~ ingress   = [
    - {
      - cidr_blocks = [
        - "0.0.0.0/0",
      ]
      - from_port   = 22
      - ipv6_cidr_blocks = []
      - prefix_list_ids = []
      - protocol     = "tcp"
      - security_groups = []
      - self         = false
      - to_port      = 22
      # (1 unchanged attribute hidden)
    },
    - {
      - cidr_blocks = [
        - "0.0.0.0/0",
      ]
    },
  ]
}

```

Apply complete! Resources: 0 added, 1 changed, 0 destroyed.

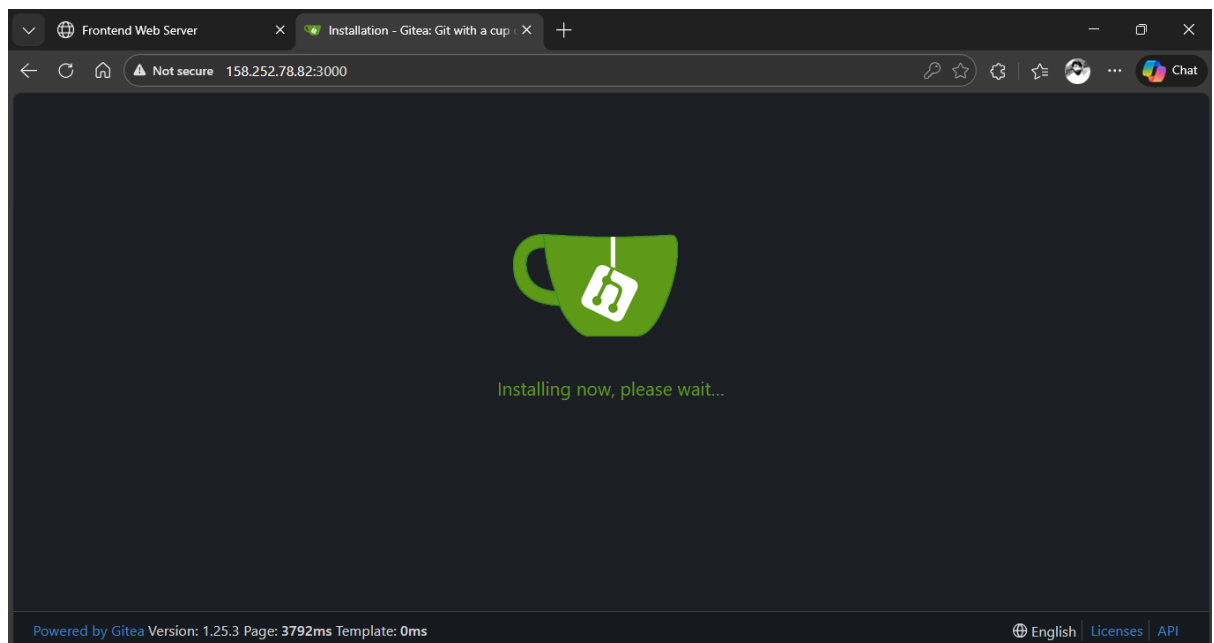
Outputs:

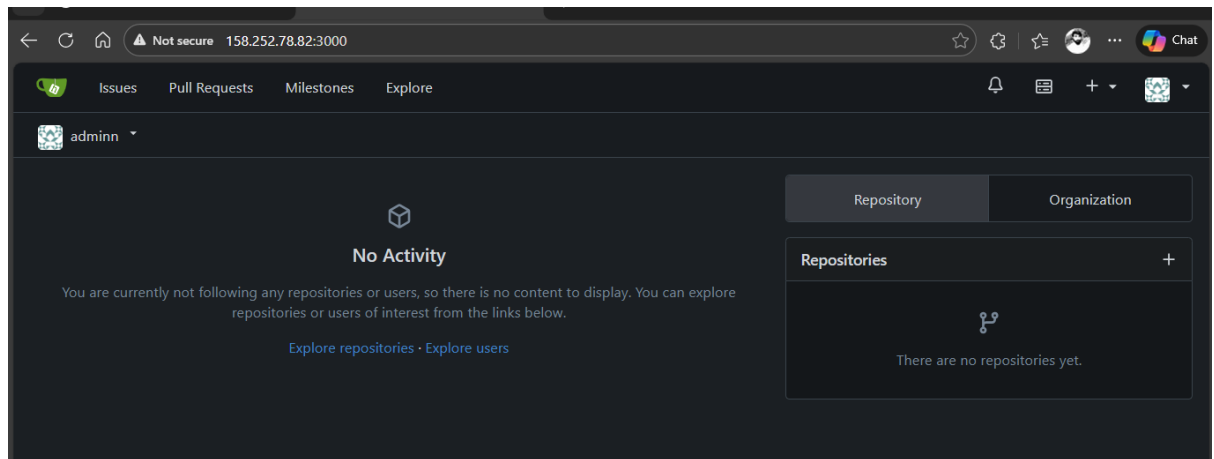
```

webserver_public_ips = [
  "158.252.78.82",
]

```

- task9_gitea_browser.png





📌 Task 10 – Terraform + Ansible Automation (null_resource)

Tasks

- Add null_resource
- Trigger Ansible from Terraform
- Fix SSH readiness with wait task

Screenshots

- task10_null_resource_main_tf.png

```
resource "null_resource" "configure_server" {
  triggers = {
    webserver_public_ips_for_ansible = module.myapp-webserver[0].aws_instance_public_ip
  }

  depends_on = [module.myapp-webserver]

  provisioner "local-exec" {
    command = <<-EOT
      ansible-playbook -i ${self.triggers.webserver_public_ips_for_ansible}, \
        --private-key ~/.ssh/id_ed25519 --user ec2-user \
        my-playbook.yaml
    EOT
  }
}
```

- task10_terraform_destroy_before_null.png


```

@toobashafique065 @ /workspaces/terraform_machine (main) $ terraform destroy -auto-approve
module.myapp-webserver[0].aws_key_pair.ssh-key: Refreshing state... [id=dev-serverkey-0]
aws_vpc.myapp_vpc: Refreshing state... [id=vpc-0f313f58fd0996cd0]
module.myapp-subnet.aws_internet_gateway.myapp_igw: Refreshing state... [id=igw-0a86095ed97fd33d4]
module.myapp-subnet.aws_subnet.myapp_subnet_1: Refreshing state... [id=subnet-0a454ec6f2a1ce5]
module.myapp-webserver[0].aws_security_group.web_sg: Refreshing state... [id=sg-0bf285c4870ec7dff]
module.myapp-subnet.aws_default_route_table.main_rt: Refreshing state... [id=rtb-0d1f69a2073f9fd17]
module.myapp-webserver[0].aws_instance.myapp-server: Refreshing state... [id=i-01d7b80f722d4f566]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
- destroy

Terraform will perform the following actions:

# aws_vpc.myapp_vpc will be destroyed
- resource "aws_vpc" "myapp_vpc" {
  - arn                                = "arn:aws:ec2:me-central-1:535942231766:vpc/vpc-0f313f58fd0996cd0" -> null
  - assign_generated_ipv6_cidr_block   = false -> null
  - cidr_block                         = "10.0.0.0/16" -> null
  - default_network_acl_id            = "acl-0eb7230b55259fd9d" -> null
  - default_route_table_id            = "rtb-0d1f69a2073f9fd17" -> null
  - default_security_group_id         = "sg-062bcb28cec0a999d" -> null
  - dhcp_options_id                   = "dopt-02c0a1e8f3dd284d6" -> null
  - enable_dns_hostnames               = false -> null
  - enable_dns_support                = true -> null
  - enable_network_address_usage_metrics = false -> null
  - id                                = "vpc-0f313f58fd0996cd0" -> null
  - instance_tenancy                  = "default" -> null
  - ipv6_netmask_length                = 0 -> null
  - main_route_table_id                = "rtb-0d1f69a2073f9fd17" -> null
  - owner_id                           = "535942231766" -> null
  - region                             = "me-central-1" -> null
  - tags                               = {
    - "Name" = "dev-vpc"
  } -> null
  - tags_all                           = {
    - "Name" = "dev-vpc"
  } -> null
  # (4 unchanged attributes hidden)

```

- task10_my_playbook_wait_for_ssh.png

```

- name: Wait for some time to ensure system readiness
  hosts: all
  tasks:
    - name: Wait 300 seconds for SSH to become ready
      wait_for:
        port: 22
        host: "{{ inventory_hostname }}"
        delay: 10
        timeout: 300
      delegate_to: localhost

```

- task10_terraform_apply_after_wait.png

```

gtoobashafique065 @ /workspaces/terraform_machine (main) $ terraform apply
module.myapp-webserver[0].aws_key_pair.ssh-key: Refreshing state... [id=dev-serverkey-0]
aws_vpc.myapp_vpc: Refreshing state... [id=vpc-0205617480dd13552]
module.myapp-subnet.aws_internet_gateway.myapp_igw: Refreshing state... [id=igw-0804e2e546734638f]
module.myapp-subnet.aws_subnet.myapp_subnet_1: Refreshing state... [id=subnet-06457d2c9d8e6bde8]
module.myapp-webserver[0].aws_security_group.web_sg: Refreshing state... [id=sg-08f82ea6da5036e6c]
module.myapp-subnet.aws_default_route_table.main_rt: Refreshing state... [id=rtb-0674138a5d65be459]
module.myapp-webserver[0].aws_instance.myapp-server: Refreshing state... [id=i-09d4eabad356e65ec]
null_resource.configure_server: Refreshing state... [id=8789740229408547428]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the fol
-/+ destroy and then create replacement

Terraform will perform the following actions:

  # null_resource.configure_server is tainted, so must be replaced
  -/+ resource "null_resource" "configure_server" {
    ~ id      = "8789740229408547428" -> (known after apply)
    # (1 unchanged attribute hidden)
  }

Plan: 1 to add, 0 to change, 1 to destroy.

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes

null_resource.configure_server: Destroying... [id=8789740229408547428]
null_resource.configure_server: Destruction complete after 0s
null_resource.configure_server: Creating...
null_resource.configure_server: Provisioning with 'local-exec'...
null_resource.configure_server (local-exec): Executing: ["/bin/sh" "-c" "sleep 60\nANSIBLE_HOST_KEY_CHECKING=False ansible-
ate-key ~/.ssh/id_ed25519 \\n-u ec2-user \\nmy-playbook.yaml\n"]

```

- task10_app_browser_post_null_resource.png

40.172.100.31 Installation - Gitea: Git with a cup of ☕

Not secure 40.172.100.31:3000

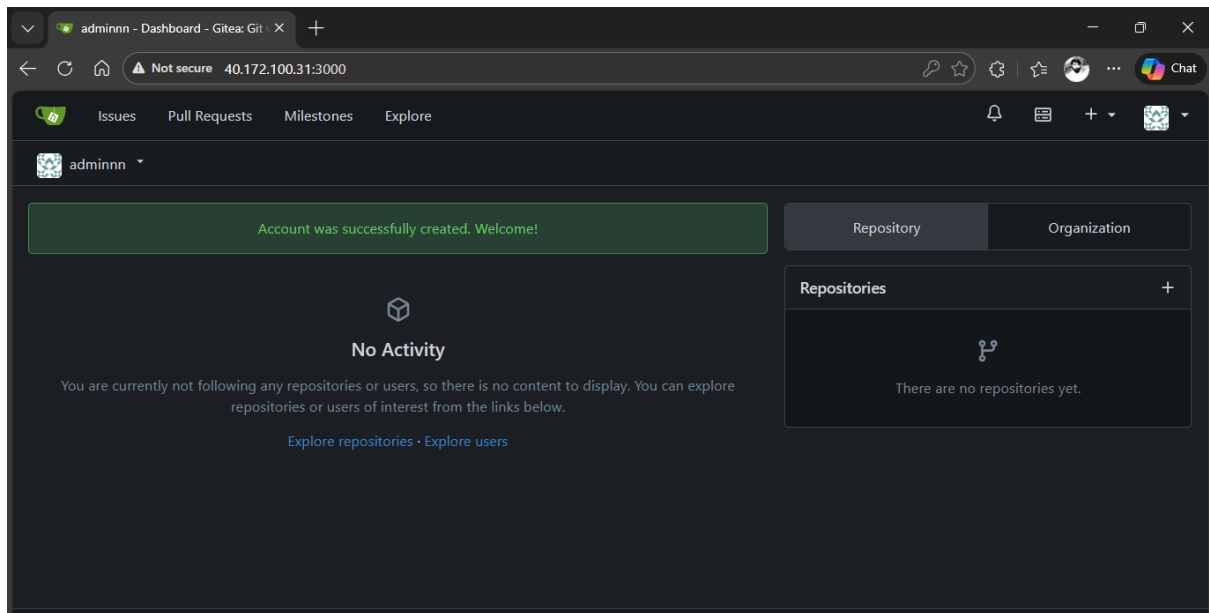
Initial Configuration

If you run Gitea inside Docker, please read the [documentation](#) before changing any settings.

Database Settings

Gitea requires MySQL, PostgreSQL, MSSQL, SQLite3 or TiDB (MySQL protocol).

Database Type *	PostgreSQL
Host *	db:5432
Username *	gitea
Password *	
Database Name *	gitea
SSL *	Disable
Schema	



📌 Task 11 – Dynamic Inventory (aws_ec2 Plugin)

Tasks

- Configure aws_ec2 plugin
- Create dynamic inventory
- Deploy dev & prod EC2 instances

Screenshots

- task11_ansible_cfg_aws_ec2.png

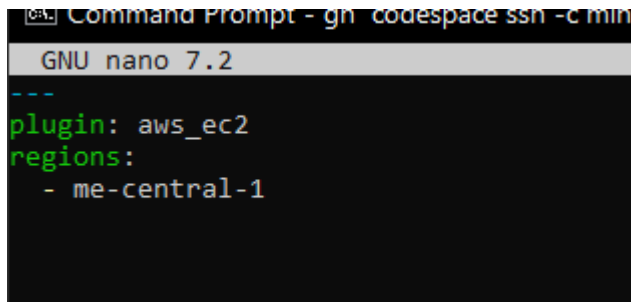
```
Command Prompt - gh codespace ssh -c miniature-
GNU nano 7.2
[defaults]
host_key_checking = False
interpreter_python = /usr/bin/python3
deprecation_warnings = False

enable_plugins = aws_ec2
private_key_file = ~/.ssh/id_ed25519
```

- task11_inventory_aws_ec2_created.png

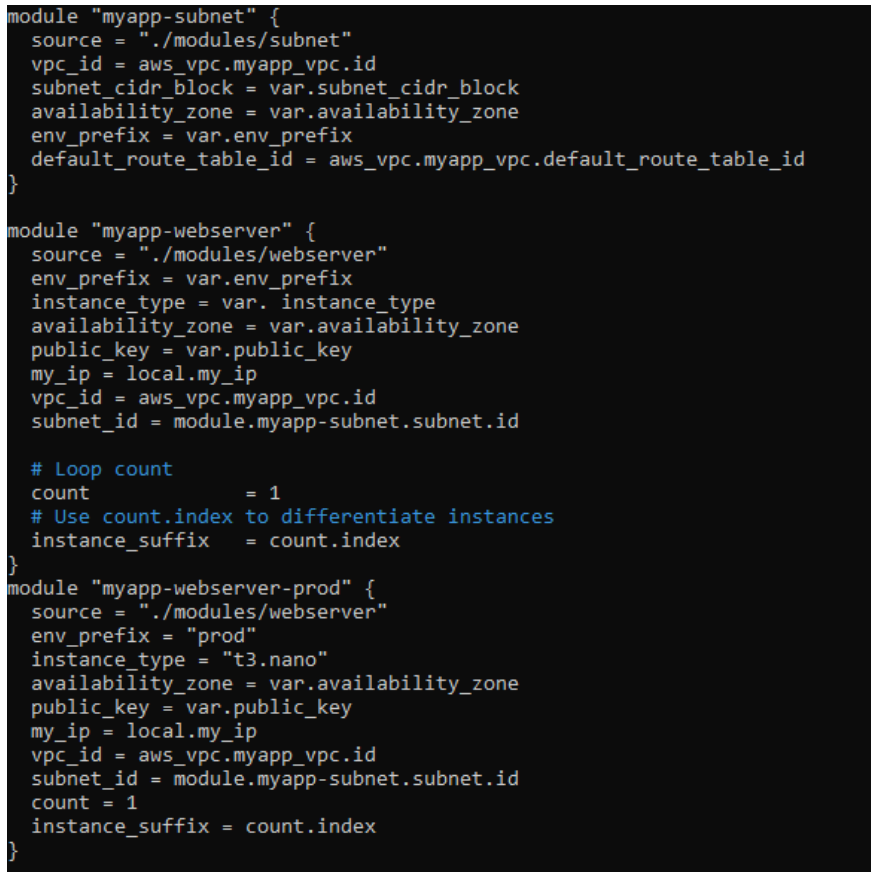
```
@toobashafique065 /workspaces/terraform_machine (main) $ ls -la inventory_aws_ec2.yaml
-rw-rw-rw- 1 codespace codespace 0 Jan 10 19:46 inventory_aws_ec2.yaml
@toobashafique065 /workspaces/terraform_machine (main) $
```

- task11_inventory_aws_ec2_initial.png



```
Command Prompt - gh codespace ssh -c min
GNU nano 7.2
---
plugin: aws_ec2
regions:
  - me-central-1
```

- task11_main_tf_dev_prod_modules.png



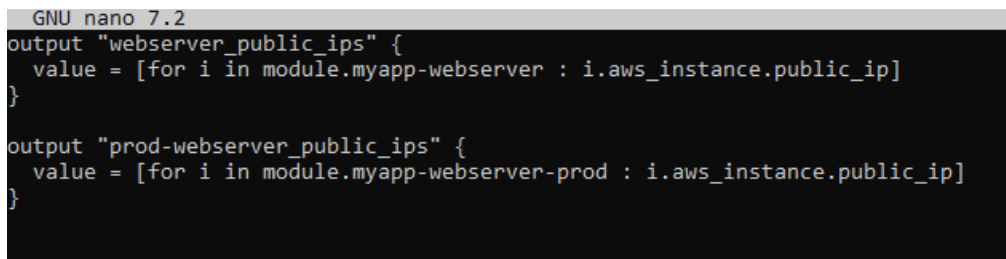
```
module "myapp-subnet" {
  source = "./modules/subnet"
  vpc_id = aws_vpc.myapp_vpc.id
  subnet_cidr_block = var.subnet_cidr_block
  availability_zone = var.availability_zone
  env_prefix = var.env_prefix
  default_route_table_id = aws_vpc.myapp_vpc.default_route_table_id
}

module "myapp-webserver" {
  source = "./modules/webserver"
  env_prefix = var.env_prefix
  instance_type = var.instance_type
  availability_zone = var.availability_zone
  public_key = var.public_key
  my_ip = local.my_ip
  vpc_id = aws_vpc.myapp_vpc.id
  subnet_id = module.myapp-subnet.subnet.id

  # Loop count
  count = 1
  # Use count.index to differentiate instances
  instance_suffix = count.index
}

module "myapp-webserver-prod" {
  source = "./modules/webserver"
  env_prefix = "prod"
  instance_type = "t3.nano"
  availability_zone = var.availability_zone
  public_key = var.public_key
  my_ip = local.my_ip
  vpc_id = aws_vpc.myapp_vpc.id
  subnet_id = module.myapp-subnet.subnet.id
  count = 1
  instance_suffix = count.index
}
```

- task11_outputs_tf_dev_prod_ips.png



```
GNU nano 7.2
output "webserver_public_ips" {
  value = [for i in module.myapp-webserver : i.aws_instance.public_ip]
}

output "prod-webserver_public_ips" {
  value = [for i in module.myapp-webserver-prod : i.aws_instance.public_ip]
}
```

- task11_terraform_apply_dynamic_setup.png

```
commands will detect it and remind you to do so if necessary.
@toobashafique065 @ /workspaces/terraform_machine (main) $ terraform apply -auto-approve
null_resource.configure_server: Refreshing state... [id=6944963618505235679]
module.myapp-webserver[0].aws_key_pair.ssh-key: Refreshing state... [id=dev-serverkey-0]
aws_vpc.myapp_vpc: Refreshing state... [id=vpc-0205617480dd13552]
module.myapp-subnet.aws_subnet.myapp_subnet_1: Refreshing state... [id=subnet-06457d2c9d8e6bde8]
module.myapp-subnet.aws_internet_gateway.myapp_igw: Refreshing state... [id=igw-0804e2e546734638f]
module.myapp-webserver[0].aws_security_group.web_sg: Refreshing state... [id=sg-08f82ea6da5036e6c]
module.myapp-subnet.aws_default_route_table.main_rt: Refreshing state... [id=rtb-0674138a5d65be459]
module.myapp-webserver[0].aws_instance.myapp-server: Refreshing state... [id=i-09d4eabad356e5ec]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create
~ update in-place
- destroy

Terraform will perform the following actions:

# aws_vpc.myapp_vpc will be updated in-place
~ resource "aws_vpc" "myapp_vpc" {
  ~ enable_dns_hostnames = false -> true
    id                   = "vpc-0205617480dd13552"
    tags                 = {
      "Name" = "dev-vpc"
    }
  # (19 unchanged attributes hidden)
}

# null_resource.configure_server will be destroyed
# (because null_resource.configure_server is not in configuration)
- resource "null_resource" "configure_server" {
  - id       = "6944963618505235679" -> null
  - triggers = {
    - "webserver_ip" = "40.172.100.31"
  } -> null
}

# module.myapp-webserver-prod[0].aws_instance.myapp-server will be created
+ resource "aws_instance" "myapp-server" {
  + ami = "ami-05524d6658fcf35b6"
  + arn = (known after apply)
}
```

- task11_terraform_output_dynamic_ips.png

```
Outputs:
prod-webserver_public_ips = [
  "3.29.113.101",
]
webserver_public_ips = [
  "40.172.100.31",
]
```

- task11_boto_install.png

```
@toobashafique065 @ /workspaces/terraform_machine (main) $ $(which python) -m pip install boto3 botocore
Collecting boto3
  Downloading boto3-1.42.25-py3-none-any.whl.metadata (6.8 kB)
Collecting botocore
  Downloading botocore-1.42.25-py3-none-any.whl.metadata (5.9 kB)
Collecting jmespath<2.0.0,>=0.7.1 (from boto3)
  Downloading jmespath-1.0.1-py3-none-any.whl.metadata (7.6 kB)
Collecting s3transfer<0.17.0,>=0.16.0 (from boto3)
  Downloading s3transfer-0.16.0-py3-none-any.whl.metadata (1.7 kB)
Requirement already satisfied: python-dateutil<3.0.0,>=2.1 in /home/codespace/.local/lib/python3.12/site-packages (from botocore) (2.9.0.post0)
Requirement already satisfied: urllib3!=2.2.0,<3,>=1.25.4 in /home/codespace/.local/lib/python3.12/site-packages (from botocore) (2.5.0)
Requirement already satisfied: six>=1.5 in /home/codespace/.local/lib/python3.12/site-packages (from python-dateutil<3.0.0,>=2.1->botocore) (1.17.0)
Downloading boto3-1.42.25-py3-none-any.whl (140 kB)
Downloading botocore-1.42.25-py3-none-any.whl (14.6 MB)
 14.6/14.6 MB 27.4 MB/s 0:00:00
Downloading jmespath-1.0.1-py3-none-any.whl (20 kB)
Downloading s3transfer-0.16.0-py3-none-any.whl (86 kB)
Installing collected packages: jmespath, botocore, s3transfer, boto3
Successfully installed boto3-1.42.25 botocore-1.42.25 jmespath-1.0.1 s3transfer-0.16.0
```

- task11_boto_version.png

```
@toobashafique065 @ /workspaces/terraform_machine (main) $ $(which python) -c "import boto3, botocore; print(boto3.__version__)"
1.42.25
```

- task11_ansible_inventory_graph_initial.png

```
@toobashafique065 [ ] /workspaces/terraform_machine (main) $ ansible-inventory -i inventory_aws_ec2.yaml --graph
[WARNING]: Ansible is being run in a world writable directory (/workspaces/terraform_machine), ignoring it as an ans
cs.ansible.com/ansible/devel/reference_appendices/config.html#cfg-in-world-writable-dir
[WARNING]: Deprecation warnings can be disabled by setting 'deprecation_warnings=False' in ansible.cfg.
[DEPRECATION WARNING]: Importing 'to_text' from 'ansible.module_utils._text' is deprecated. This feature will be re
ule_utils.common.text.converters instead.
[DEPRECATION WARNING]: Importing 'to_native' from 'ansible.module_utils._text' is deprecated. This feature will be
odule_utils.common.text.converters instead.
[DEPRECATION WARNING]: Passing 'disable_lookups' to 'template' is deprecated. This feature will be removed from ansi
@all:
|--@ungrouped:
|--@aws_ec2:
|   |--ec2-40-172-100-31.me-central-1.compute.amazonaws.com
|   |--ec2-3-29-113-101.me-central-1.compute.amazonaws.com
```

📌 Task 12 – Filtering by Tags & Instance Type

Tasks

- Group EC2s by tags and instance type
- Run Ansible on filtered groups

Screenshots

- task12_inventory_aws_ec2_tag_groups.png

```
---
plugin: aws_ec2
regions:
- me-central-1
keyed_groups:
- key: tags
  prefix: tag
  separator: "_"
```

- task12_inventory_graph_tag_groups.png

```
@toobashafique065 [ ] /workspaces/terraform_machine (main) $ ansible-inventory -i inventory_aws_ec2.yaml --graph
[WARNING]: Ansible is being run in a world writable directory (/workspaces/terraform_machine), ignoring it as an ans
cs.ansible.com/ansible/devel/reference_appendices/config.html#cfg-in-world-writable-dir
[WARNING]: Deprecation warnings can be disabled by setting 'deprecation_warnings=False' in ansible.cfg.
[DEPRECATION WARNING]: Importing 'to_text' from 'ansible.module_utils._text' is deprecated. This feature will be re
ule_utils.common.text.converters instead.
[DEPRECATION WARNING]: Importing 'to_native' from 'ansible.module_utils._text' is deprecated. This feature will be r
odule_utils.common.text.converters instead.
[DEPRECATION WARNING]: Passing 'disable_lookups' to 'template' is deprecated. This feature will be removed from ansi
@all:
|--@ungrouped:
|--@aws_ec2:
|   |--ec2-40-172-100-31.me-central-1.compute.amazonaws.com
|   |--ec2-3-29-113-101.me-central-1.compute.amazonaws.com
|--@tag_Name_dev_ec2_instance_0:
|   |--ec2-40-172-100-31.me-central-1.compute.amazonaws.com
|--@tag_Name_prod_ec2_instance_0:
|   |--ec2-3-29-113-101.me-central-1.compute.amazonaws.com
```

- task12_inventory_aws_ec2_instance_type_groups.png

```

---
plugin: aws_ec2
regions:
  - me-central-1
keyed_groups:
  - key: tags
    prefix: tag
    separator: "-"
  - key: instance_type
    prefix: instance_type
    separator: "-"

```

- task12_inventory_graph_full.png

```

@toobashafique065 [ /workspaces/terraform_machine (main) ] $ ansible-inventory -i inventory_aws_ec2.yaml --graph
[WARNING]: Ansible is being run in a world writable directory (/workspaces/terraform_machine), ignoring it as an a
cs.ansible.com/ansible/devel/reference_appendices/config.html#cfg-in-world-writable-dir
[WARNING]: Deprecation warnings can be disabled by setting `deprecation_warnings=False` in ansible.cfg.
[DEPRECATION WARNING]: Importing 'to_text' from 'ansible.module_utils._text' is deprecated. This feature will be r
ule_utils.common.text.converters instead.
[DEPRECATION WARNING]: Importing 'to_native' from 'ansible.module_utils._text' is deprecated. This feature will be
odule_utils.common.text.converters instead.
[DEPRECATION WARNING]: Passing `disable_lookups` to `template` is deprecated. This feature will be removed from an
@all:
|--@ungrouped:
|--@aws_ec2:
|   |--ec2-40-172-100-31.me-central-1.compute.amazonaws.com
|   |--ec2-3-29-113-101.me-central-1.compute.amazonaws.com
|--@tag_Name_dev_ec2_instance_0:
|   |--ec2-40-172-100-31.me-central-1.compute.amazonaws.com
|--@instance_type_t3_micro:
|   |--ec2-40-172-100-31.me-central-1.compute.amazonaws.com
|--@tag_Name_prod_ec2_instance_0:
|   |--ec2-3-29-113-101.me-central-1.compute.amazonaws.com
|--@instance_type_t3_nano:
|   |--ec2-3-29-113-101.me-central-1.compute.amazonaws.com
@toobashafique065 [ /workspaces/terraform_machine (main) ] $

```

- task12_my_playbook_all_hosts.png

```
---
- name: Configure nginx web server
  hosts: all
  become: true
  tasks:
    - name: install nginx and update cache
      yum:
        name: nginx
        state: present
        update_cache: yes

    - name: install openssl
      yum:
        name: openssl
        state: present

    - name: start nginx server
      service:
        name: nginx
        state: started
        enabled: true

- name: Configure SSL certificates
  hosts: all
  become: true
  tasks:
    - name: Create SSL private directory
      file:
        path: /etc/ssl/private
        state: directory
        mode: '0700'

    - name: Create SSL certs directory
      file:
        path: /etc/ssl/certs
        state: directory
        mode: '0755'

    - name: Get IMDSv2 token
      uri:
```

- task12_ansible_play_all.png


```

@roobashafique065 @ /workspaces/terraform_machine (main) $ ansible-playbook my-playbook.yaml

PLAY [Configure nginx web server] *****
[WARNING]: Found variable using reserved name 'tags'.
Origin: <unknown>

tags

TASK [Gathering Facts] *****
ok: [ec2-3-29-113-101.me-central-1.compute.amazonaws.com]
ok: [ec2-40-172-100-31.me-central-1.compute.amazonaws.com]

TASK [install nginx and update cache] *****
changed: [ec2-40-172-100-31.me-central-1.compute.amazonaws.com]
changed: [ec2-3-29-113-101.me-central-1.compute.amazonaws.com]

TASK [install openssl] *****
ok: [ec2-40-172-100-31.me-central-1.compute.amazonaws.com]
ok: [ec2-3-29-113-101.me-central-1.compute.amazonaws.com]

TASK [start nginx server] *****
changed: [ec2-40-172-100-31.me-central-1.compute.amazonaws.com]
changed: [ec2-3-29-113-101.me-central-1.compute.amazonaws.com]

PLAY [Configure SSL certificates] *****

TASK [Gathering Facts] *****
ok: [ec2-40-172-100-31.me-central-1.compute.amazonaws.com]
ok: [ec2-3-29-113-101.me-central-1.compute.amazonaws.com]

TASK [Create SSL private directory] *****
changed: [ec2-40-172-100-31.me-central-1.compute.amazonaws.com]
changed: [ec2-3-29-113-101.me-central-1.compute.amazonaws.com]

TASK [Create SSL certs directory] *****

```

- task12_ansible_play_dev_only.png

```

@roobashafique065 @ /workspaces/terraform_machine (main) $ ansible-playbook -l tag_Name_dev_* my-playbook.yaml

PLAY [Configure nginx web server] *****
[WARNING]: Found variable using reserved name 'tags'.
Origin: <unknown>

tags

TASK [Gathering Facts] *****
ok: [ec2-40-172-100-31.me-central-1.compute.amazonaws.com]

TASK [install nginx and update cache] *****
ok: [ec2-40-172-100-31.me-central-1.compute.amazonaws.com]

TASK [install openssl] *****
ok: [ec2-40-172-100-31.me-central-1.compute.amazonaws.com]

TASK [start nginx server] *****
ok: [ec2-40-172-100-31.me-central-1.compute.amazonaws.com]

PLAY [Configure SSL certificates] *****

TASK [Gathering Facts] *****
ok: [ec2-40-172-100-31.me-central-1.compute.amazonaws.com]

TASK [Create SSL private directory] *****
ok: [ec2-40-172-100-31.me-central-1.compute.amazonaws.com]

TASK [Create SSL certs directory] *****
ok: [ec2-40-172-100-31.me-central-1.compute.amazonaws.com]

TASK [Get IMDSv2 token] *****
ok: [ec2-40-172-100-31.me-central-1.compute.amazonaws.com]

TASK [Get current public IP] *****
ok: [ec2-40-172-100-31.me-central-1.compute.amazonaws.com]

```

- task12_ansible_play_prod_only.png

```

@toobashafique065 [ /workspaces/terraform_machine (main) ] $ ansible-playbook -l tag_Name_prod_* my-playbook.yaml

PLAY [Configure nginx web server] *****
[WARNING]: Found variable using reserved name 'tags'.
Origin: <unknown>

tags

TASK [Gathering Facts] *****
ok: [ec2-3-29-113-101.me-central-1.compute.amazonaws.com]

TASK [install nginx and update cache] *****
ok: [ec2-3-29-113-101.me-central-1.compute.amazonaws.com]

TASK [install openssl] *****
ok: [ec2-3-29-113-101.me-central-1.compute.amazonaws.com]

TASK [start nginx server] *****
ok: [ec2-3-29-113-101.me-central-1.compute.amazonaws.com]

PLAY [Configure SSL certificates] *****

TASK [Gathering Facts] *****
ok: [ec2-3-29-113-101.me-central-1.compute.amazonaws.com]

TASK [Create SSL private directory] *****
ok: [ec2-3-29-113-101.me-central-1.compute.amazonaws.com]

TASK [Create SSL certs directory] *****
ok: [ec2-3-29-113-101.me-central-1.compute.amazonaws.com]

TASK [Get IMDSv2 token] *****

```

- task12_ansible_play_t3_micro.png

```

@toobashafique065 [ /workspaces/terraform_machine (main) ] $ ansible-playbook -l instance_type_t3_micro my-playbook.yaml

PLAY [Configure nginx web server] *****
[WARNING]: Found variable using reserved name 'tags'.
Origin: <unknown>

tags

TASK [Gathering Facts] *****
ok: [ec2-40-172-100-31.me-central-1.compute.amazonaws.com]

TASK [install nginx and update cache] *****
ok: [ec2-40-172-100-31.me-central-1.compute.amazonaws.com]

TASK [install openssl] *****
ok: [ec2-40-172-100-31.me-central-1.compute.amazonaws.com]

TASK [start nginx server] *****
ok: [ec2-40-172-100-31.me-central-1.compute.amazonaws.com]

PLAY [Configure SSL certificates] *****

TASK [Gathering Facts] *****
ok: [ec2-40-172-100-31.me-central-1.compute.amazonaws.com]

TASK [Create SSL private directory] *****
ok: [ec2-40-172-100-31.me-central-1.compute.amazonaws.com]

TASK [Create SSL certs directory] *****
ok: [ec2-40-172-100-31.me-central-1.compute.amazonaws.com]

TASK [Get IMDSv2 token] *****
ok: [ec2-40-172-100-31.me-central-1.compute.amazonaws.com]

TASK [Get current public IP] *****
ok: [ec2-40-172-100-31.me-central-1.compute.amazonaws.com]

TASK [Show current public IP] *****

```

- task12_ansible_play_t3_nano.png

```

@roobashafique065 @ /workspaces/terraform_machine (main) $ ansible-playbook -l instance_type_t3_nano my-playbook.yaml

PLAY [Configure nginx web server] *****
[WARNING]: Found variable using reserved name 'tags'.
Origin: <unknown>

tags

TASK [Gathering Facts] *****
ok: [ec2-3-29-113-101.me-central-1.compute.amazonaws.com]

TASK [install nginx and update cache] *****
ok: [ec2-3-29-113-101.me-central-1.compute.amazonaws.com]

TASK [install openssl] *****
ok: [ec2-3-29-113-101.me-central-1.compute.amazonaws.com]

TASK [start nginx server] *****
ok: [ec2-3-29-113-101.me-central-1.compute.amazonaws.com]

PLAY [Configure SSL certificates] *****
TASK [Gathering Facts] *****
ok: [ec2-3-29-113-101.me-central-1.compute.amazonaws.com]

TASK [Create SSL private directory] *****
ok: [ec2-3-29-113-101.me-central-1.compute.amazonaws.com]

TASK [Create SSL certs directory] *****
ok: [ec2-3-29-113-101.me-central-1.compute.amazonaws.com]

TASK [Get IMDSv2 token] *****
ok: [ec2-3-29-113-101.me-central-1.compute.amazonaws.com]

TASK [Get current public IP] *****

```

- task12_ansible_cfg_inventory_default.png

```

ec2-user@ip-10-0-10-167:~
GNU nano 7.2
[defaults]
inventory = ./inventory_aws_ec2.yaml
host_key_checking = False
deprecation_warnings = False
interpreter_python = /usr/bin/python3
remote_user = ec2-user
private_key_file = ~/.ssh/id_ed25519

```

- task12_ansible_play_t3_nano_no_i.png

```

@roobashafique065 @ /workspaces/terraform_machine (main) $ ansible-playbook -l instance_type_t3_nano my-playbook.yaml

PLAY [Configure nginx web server] *****
[WARNING]: Found variable using reserved name 'tags'.
Origin: <unknown>

tags

TASK [Gathering Facts] *****
ok: [ec2-3-29-113-101.me-central-1.compute.amazonaws.com]

TASK [install nginx and update cache] *****
ok: [ec2-3-29-113-101.me-central-1.compute.amazonaws.com]

TASK [install openssl] *****
ok: [ec2-3-29-113-101.me-central-1.compute.amazonaws.com]

TASK [start nginx server] *****
ok: [ec2-3-29-113-101.me-central-1.compute.amazonaws.com]

PLAY [Configure SSL certificates] *****
TASK [Gathering Facts] *****
ok: [ec2-3-29-113-101.me-central-1.compute.amazonaws.com]

TASK [Create SSL private directory] *****
ok: [ec2-3-29-113-101.me-central-1.compute.amazonaws.com]

TASK [Create SSL certs directory] *****
ok: [ec2-3-29-113-101.me-central-1.compute.amazonaws.com]

TASK [Get IMDSv2 token] *****
ok: [ec2-3-29-113-101.me-central-1.compute.amazonaws.com]

TASK [Get current public IP] *****
ok: [ec2-3-29-113-101.me-central-1.compute.amazonaws.com]

```

Task 13 – Ansible Roles (nginx, ssl, webapp)

Tasks

- Create Ansible roles
- Convert playbooks to role-based
- Deploy nginx, SSL & webapp using roles

Screenshots

- task13_main_tf_single_dev.png

```
GNU nano 7.2
provider "aws" {
  shared_config_files      = ["~/.aws/config"]
  shared_credentials_files = ["~/.aws/credentials"]
}

resource "aws_vpc" "myapp_vpc" {
  cidr_block           = var.vpc_cidr_block
  enable_dns_hostnames = true
  tags = {
    Name = "${var.env_prefix}-vpc"
  }
}

module "myapp-subnet" {
  source                = "./modules/subnet"
  vpc_id                = aws_vpc.myapp_vpc.id
  subnet_cidr_block     = var.subnet_cidr_block
  availability_zone     = var.availability_zone
  env_prefix            = var.env_prefix
  default_route_table_id = aws_vpc.myapp_vpc.default_route_table_id
}

module "myapp-webserver" {
  source                = "./modules/webserver"
  env_prefix            = var.env_prefix
  instance_type         = var.instance_type
  availability_zone     = var.availability_zone
  public_key            = var.public_key
  my_ip                 = local.my_ip
  vpc_id                = aws_vpc.myapp_vpc.id
  subnet_id             = module.myapp-subnet.subnet.id

  count                = 1
  instance_suffix      = count.index
}
```

- task13_ansible_structure_created.png

```
Last login: Thu Jan 15 17:48:55 2026 from ::1
@toobashafique065 [ ] /workspaces/terraform_machine (main) $ > main.tf
@toobashafique065 [ ] /workspaces/terraform_machine (main) $ nano main.tf
@toobashafique065 [ ] /workspaces/terraform_machine (main) $ mkdir -p ansible/inventory roles
@toobashafique065 [ ] /workspaces/terraform_machine (main) $ touch ansible/ansible.cfg ansible/my-playbook.yaml
@toobashafique065 [ ] /workspaces/terraform_machine (main) $ ls -R ansible
ansible:
ansible.cfg  inventory  my-playbook.yaml
ansible/inventory:
```

- task13_ansible_cfg_project.png

```
GNU nano 7.2
[defaults]
host_key_checking = False
interpreter_python = /usr/bin/python3
```

- task13_ansible_inventory_hosts.png

```
GNU nano 7.2
[nginx]
40.172.100.31

[nginx:vars]
ansible_ssh_private_key_file=~/.ssh/id_ed25519
ansible_user=ec2-user
```

- task13_roles_created.png

```
@toobashafique065 @ /workspaces/terraform_machine (main) $ cd ansible/roles
@toobashafique065 @ /workspaces/terraform_machine/ansible/roles (main) $ ansible-galaxy role init nginx
- Role nginx was created successfully
@toobashafique065 @ /workspaces/terraform_machine/ansible/roles (main) $ ansible-galaxy role init ssl
- Role ssl was created successfully
@toobashafique065 @ /workspaces/terraform_machine/ansible/roles (main) $ ansible-galaxy role init webapp
- Role webapp was created successfully
@toobashafique065 @ /workspaces/terraform_machine/ansible/roles (main) $ cd ..
@toobashafique065 @ /workspaces/terraform_machine/ansible (main) $ ls -R roles
roles:
nginx  ssl  webapp

roles/nginx:
README.md  defaults  files  handlers  meta  tasks  templates  tests  vars

roles/nginx/defaults:
main.yml

roles/nginx/files:

roles/nginx/handlers:
main.yml

roles/nginx/meta:
main.yml

roles/nginx/tasks:
main.yml

roles/nginx/templates:

roles/nginx/tests:
inventory  test.yml

roles/nginx/vars:
main.yml

roles/ssl:
README.md  defaults  files  handlers  meta  tasks  templates  tests  vars
```

- task13_nginx_handlers_main.png

```
GNU nano 7.2
#SPDX-License-Identifier: MIT-0
---
# handlers file for nginx
- name: Restart nginx
  service:
    name: nginx
    state: restarted
```

- task13_nginx_tasks_main.png

```
- name: Install nginx
  yum:
    name: nginx
    state: present
    update_cache: yes
  notify: Restart nginx

- name: Install openssl
  yum:
    name: openssl
    state: present

- name: Start and enable nginx
  service:
    name: nginx
    state: started
    enabled: true
```

- task13_my_playbook_nginx_only.png

```
---
- name: Deploy NGINX Web Stack
  hosts: nginx
  become: true
  roles:
    - nginx
```

- task13_ansible_play_nginx_only.png

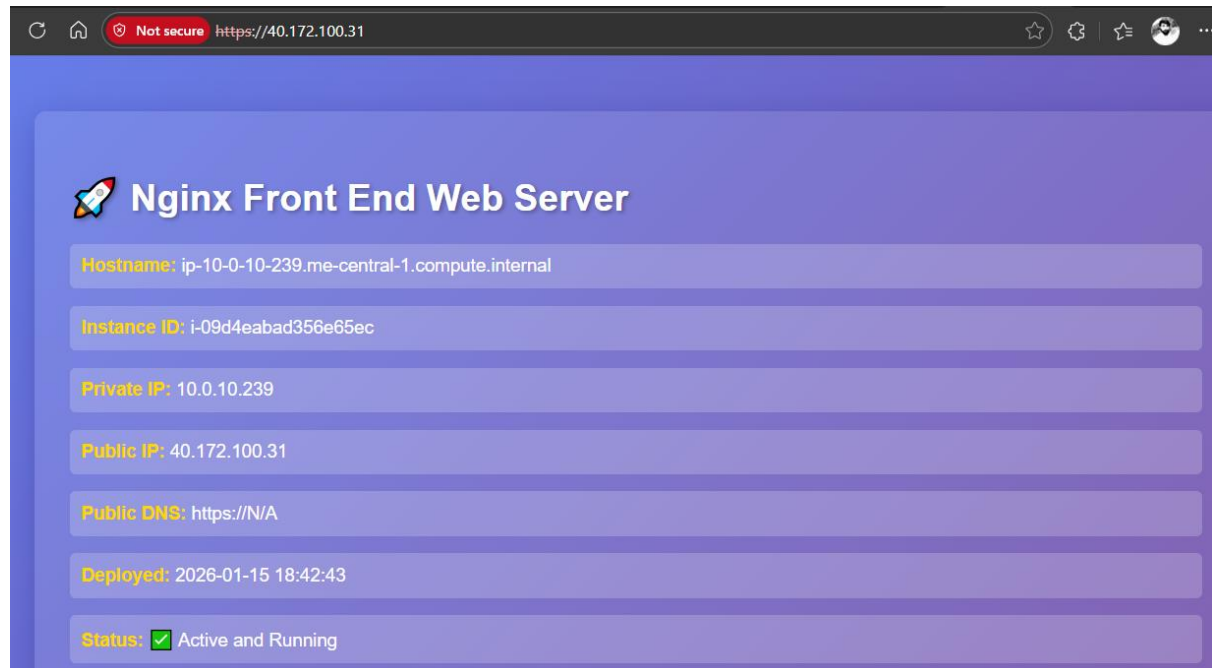
```

@toobashaFique065 @ /workspaces/terraform_machine/ansible (main) $ ansible-playbook -i inventory/hosts my-playbook.yaml

PLAY [Deploy NGINX Web Stack] *****
TASK [Gathering Facts] *****
ok: [40.172.100.31]
TASK [nginx : Install nginx] *****
ok: [40.172.100.31]
TASK [nginx : Install openssl] *****
ok: [40.172.100.31]
TASK [nginx : Start and enable nginx] *****
ok: [40.172.100.31]
PLAY RECAP *****
40.172.100.31      : ok=4   changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

```

- task13_nginx_browser_roles.png



- task13_ssl_defaults_main.png

```

GNU nano 7.2
#SPDX-License-Identifier: MIT-0
---
# defaults file for ssl
imds_v2_token_ttl: "3600"
ssl_days_valid: 365

```

- task13_ssl_tasks_main.png

```

mode: '0700'

- name: Create SSL certs directory
  file:
    path: /etc/ssl/certs
    state: directory
    mode: '0755'

- name: Get IMDSv2 token
  uri:
    url: http://169.254.169.254/latest/api/token
    method: PUT
    headers:
      X-aws-ec2-metadata-token-ttl-seconds: "{{ imdsv2_token_ttl }}"
    return_content: yes
  register: imds_token

- name: Get public IP
  uri:
    url: http://169.254.169.254/latest/meta-data/public-ipv4
    headers:
      X-aws-ec2-metadata-token: "{{ imds_token.content }}"
    return_content: yes
  register: public_ip

- name: Save public IP as fact
  set_fact:
    server_public_ip: "{{ public_ip.content }}"

- name: Generate self-signed certificate
  command: >
    openssl req -x509 -nodes -days {{ ssl_days_valid }}
    -newkey rsa:2048
    -keyout /etc/ssl/private/selfsigned.key
    -out /etc/ssl/certs/selfsigned.crt
    -subj "/CN={{ server_public_ip }}"
    -addext "subjectAltName=IP:{{ server_public_ip }}"
  args:
    creates: /etc/ssl/certs/selfsigned.crt

```

- task13_webapp_defaults_main.png

```

GNU nano 7.2
#SPDX-License-Identifier: MIT-0
---
# defaults file for webapp
nginx_user: nginx
nginx_worker_processes: auto
nginx_worker_connections: 1024
nginx_error_log_level: notice

# Webapp settings
web_root: /usr/share/nginx/html
web_index_file: index.php

```


- task13_webapp_files_index_php.png

```

GNU nano 7.2 roles/webapp/files
<?php
// Get hostname
$hostname = gethostname();

// Deployment date
$deployed_date = date("Y-m-d H:i:s");

// Metadata base URL
$metadata_base = "http://169.254.169.254/latest/";

// Function to get IMDSv2 token
function getImdsV2Token() {
    $ch = curl_init("http://169.254.169.254/latest/api/token");
    curl_setopt_array($ch, [
        CURLOPT_RETURNTRANSFER => true,
        CURLOPT_CUSTOMREQUEST => "PUT",
        CURLOPT_HTTPHEADER => [
            "X-aws-ec2-metadata-token-ttl-seconds: 21600"
        ],
        CURLOPT_TIMEOUT => 2
    ]);

    $token = curl_exec($ch);
    curl_close($ch);

    return $token ?: null;
}

// Function to fetch metadata using token
function getMetadata($path, $token) {
    $url = "http://169.254.169.254/latest/meta-data/" . $path;

    $ch = curl_init($url);
    curl_setopt_array($ch, [
        CURLOPT_RETURNTRANSFER => true,
        CURLOPT_HTTPHEADER => [
            "X-aws-ec2-metadata-token: $token"
        ],
        CURLOPT_TIMEOUT => 2
    ]);
}

```

- task13_webapp_handlers_main.png

```

Command Prompt - gh codespace ssh -c m
GNU nano 7.2
#SPDX-License-Identifier: MIT-0
---
# handlers file for webapp
- name: Restart nginx
  service:
    name: nginx
    state: restarted
- name: Restart php-fpm
  service:
    name: php-fpm
    state: restarted

```

- task13_webapp_templates_nginx_conf.png

```

user {{ nginx_user }};
worker_processes {{ nginx_worker_processes }};
error_log /var/log/nginx/error.log {{ nginx_error_log_level }};
pid /run/nginx.pid;

events {
    worker_connections {{ nginx_worker_connections }};
}

http {
    log_format main '$remote_addr - $remote_user [$time_local] "$request"'
                   '$status $body_bytes_sent "$http_referer"'
                   '"$http_user_agent" "$http_x_forwarded_for"';

    access_log /var/log/nginx/access.log main;

    sendfile        on;
    tcp_nopush      on;
    keepalive_timeout 65;
    types_hash_max_size 4096;

    include          /etc/nginx/mime.types;
    default_type     application/octet-stream;

    upstream backend_servers {
        server 158.252.94.241:80;
        server 158.252.94.242:80 backup;
    }

    server {
        listen 443 ssl;
        server_name {{ server_public_ip }};

        ssl_certificate /etc/ssl/certs/selfsigned.crt;
        ssl_certificate_key /etc/ssl/private/selfsigned.key;

        location / {
            root {{ web_root }};
            index {{ web_index_file }} index.html index.htm;
            # proxy_pass http://158.252.94.241:80;

```

- task13_webapp_tasks_main.png

```

GNU nano 7.2
#SPDX-License-Identifier: MIT-0
---
# tasks file for webapp
- name: Install PHP packages
  yum:
    name:
      - php-fpm
      - php-curl
    state: present
  notify: Restart php-fpm

- name: Copy PHP website
  copy:
    src: index.php
    dest: "{{ web_root }}/{{ web_index_file }}"
    owner: nginx
    group: nginx
    mode: '0644'
  notify: Restart nginx

- name: Deploy nginx config
  template:
    src: nginx.conf.j2
    dest: /etc/nginx/nginx.conf
  notify: Restart nginx

- name: Start and enable php-fpm
  service:
    name: php-fpm
    state: started
    enabled: true

```

- task13_my_playbook_roles_full.png

```

GNU nano 7.2
---
- name: Deploy NGINX Web Stack with SSL and PHP
  hosts: nginx
  become: true
  roles:
    - nginx
    - ssl
    - webapp

```

- task13_ansible_play_roles_full.png

```

@toobashafique065 /workspaces/terraform_machine/ansible (main) $ ansible-playbook -i inventory/hosts my-playbook.yaml

PLAY [Deploy NGINX Web Stack with SSL and PHP] *****

TASK [Gathering Facts] *****
ok: [40.172.100.31]

TASK [nginx : Install nginx] *****
ok: [40.172.100.31]

TASK [nginx : Install openssl] *****
ok: [40.172.100.31]

TASK [nginx : Start and enable nginx] *****
ok: [40.172.100.31]

TASK [ssl : Create SSL private directory] *****
ok: [40.172.100.31]

TASK [ssl : Create SSL certs directory] *****
ok: [40.172.100.31]

TASK [ssl : Get IMDSv2 token] *****
ok: [40.172.100.31]

TASK [ssl : Get public IP] *****
ok: [40.172.100.31]

TASK [ssl : Save public IP as fact] *****
ok: [40.172.100.31]

TASK [ssl : Generate self-signed certificate] *****
ok: [40.172.100.31]

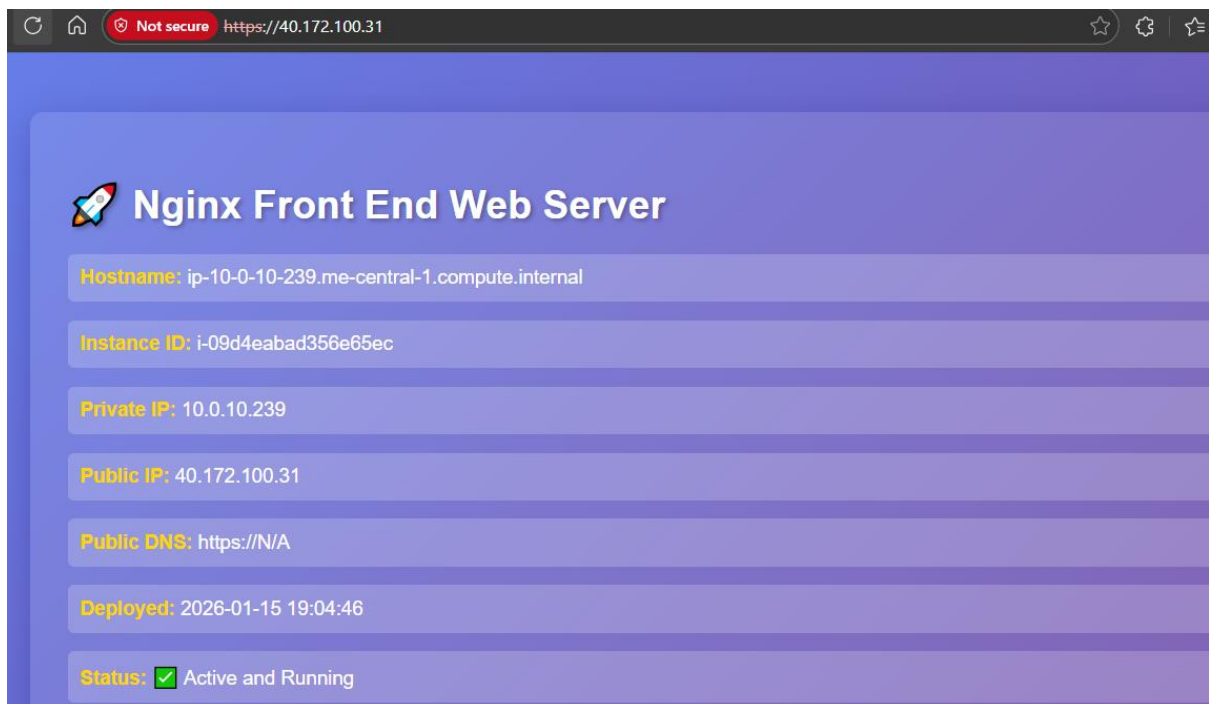
TASK [webapp : Install PHP packages] *****
ok: [40.172.100.31]

TASK [webapp : Copy PHP website] *****
changed: [40.172.100.31]

TASK [webapp : Deploy nginx config] *****
changed: [40.172.100.31]

```

- task13_php_https_browser_roles.png



Cleanup:

- cleanup_terraform_destroy.png

```

@toobashafique065 /workspaces/terraform_machine (main) $ terraform destroy -auto-approve
module.myapp-webserver[0].aws_key_pair.ssh-key: Refreshing state... [id=dev-serverkey-0]
module.myapp-webserver-prod[0].aws_key_pair.ssh-key: Refreshing state... [id=prod-serverkey-0]
aws_vpc.myapp_vpc: Refreshing state... [id=vpc-0205617480dd13552]
module.myapp-webserver-prod[0].aws_instance.myapp-server: Refreshing state... [id=i-0b6aaae3206164312]
module.myapp-webserver-prod[0].aws_security_group.web_sg: Refreshing state... [id=sg-09f5e6ef555c4d948]
module.myapp-subnet.aws_internet_gateway.myapp_igw: Refreshing state... [id=igw-0804e2e546734638f]
module.myapp-subnet.aws_subnet.myapp_subnet_1: Refreshing state... [id=subnet-06457d2c9d8e6bde8]
module.myapp-webserver[0].aws_security_group.web_sg: Refreshing state... [id=sg-08f82ea6da5036e6c]
module.myapp-subnet.aws_default_route_table.main_rt: Refreshing state... [id=rtb-0674138a5d65be459]
module.myapp-webserver[0].aws_instance.myapp-server: Refreshing state... [id=i-09d4eabad356e65ec]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
- destroy

Terraform will perform the following actions:

# aws_vpc.myapp_vpc will be destroyed
- resource "aws_vpc" "myapp_vpc" {
  - arn = "arn:aws:ec2:me-central-1:535942231766:vpc/vpc-0205617480dd13552" -> null
  - assign_generated_ipv6_cidr_block = false -> null
  - cidr_block = "10.0.0.0/16" -> null
  - default_network_acl_id = "acl-0a0bb3e84d84e2c1c" -> null
  - default_route_table_id = "rtb-0674138a5d65be459" -> null
  - default_security_group_id = "sg-0f7d4fa1a045b46b4" -> null
  - dhcp_options_id = "dopt-02c0a1e8f3dd284d6" -> null
  - enable_dns_hostnames = true -> null
  - enable_dns_support = true -> null
  - enable_network_address_usage_metrics = false -> null
  - id = "vpc-0205617480dd13552" -> null
  - instance_tenancy = "default" -> null
  - ipv6_netmask_length = 0 -> null
  - main_route_table_id = "rtb-0674138a5d65be459" -> null
  - owner_id = "535942231766" -> null
  - region = "me-central-1" -> null
  - tags = {
    - "Name" = "dev-vpc"
  } -> null
  - tags_all = {

```

- cleanup_tfstate.png

```

@toobashafique065 /workspaces/terraform_machine (main) $ cat terraform.tfstate
{
  "version": 4,
  "terraform_version": "1.14.3",
  "serial": 111,
  "lineage": "6740a468-490a-183e-b0f9-4becee99fc3e",
  "outputs": {},
  "resources": [],
  "check_results": null
}

```